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1. Dynamics of Learning in MLP: Natural Gradient and Singularity Revisited

■■■-■■■ ■■■■■, *Tomoko Ozeki, Ryo Karakida, Yuki Yoshida, Masato Okada*

https://doi.org/10.1162/neco_a_01029 | pp. 1-33

The dynamics of supervised learning play a main role in deep learning, which takes place in the parameter space of a multilayer perceptron (MLP). We review the history of supervised stochastic gradient learning, focusing on its singular structure and natural gradient. The parameter space includes singular regions in which parameters are not identifiable. One of our results is a full exploration of the dynamical behaviors of stochastic gradient learning in an elementary singular network. The bad news is its pathological nature, in which part of the singular region becomes an attractor and another part a repulser at the same time, forming a Milnor attractor. A learning trajectory is attracted by the attractor region, staying in it for a long time, before it escapes the singular region through the repulser region. This is typical of plateau phenomena in learning. We demonstrate the strange topology of a singular region by introducing blow-down coordinates, which are useful for analyzing the natural gradient dynamics. We confirm that the natural gradient dynamics are free of critical slowdown. The second main result is the good news: the interactions of elementary singular networks eliminate the attractor part and the Milnor-type attractors disappear. This explains why large-scale networks do not suffer from serious critical slowdowns due to singularities. We finally show that the unit-wise natural gradient is effective for learning in spite of its low computational cost.

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2. Balancing New against Old Information: The Role of Puzzlement Surprise in Learning

Mohammadjavad Faraji, Kerstin Preuschoff, Wulfram Gerstner

https://doi.org/10.1162/neco_a_01025 | pp. 34-83

Surprise describes a range of phenomena from unexpected events to behavioral responses. We propose a novel measure of surprise and use it for surprise-driven learning. Our surprise measure takes into account data likelihood as well as the degree of commitment to a belief via the entropy of the belief distribution. We find that surprise-minimizing learning dynamically adjusts the balance between new and old information without the need of knowledge about the temporal statistics of the environment. We apply our framework to a dynamic decision-making task and a maze exploration task. Our surprise-minimizing framework is suitable for learning in complex environments, even if the environment undergoes gradual or sudden changes, and it could eventually provide a framework to study the behavior of humans and animals as they encounter surprising events.

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3. Why Do Similarity Matching Objectives Lead to Hebbian/Anti-Hebbian Networks?

Cengiz Pehlevan, Anirvan M. Sengupta, Dmitri B. Chklovskii

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Modeling self-organization of neural networks for unsupervised learning using Hebbian and anti-Hebbian plasticity has a long history in neuroscience. Yet derivations of single-layer networks with such local learning rules from principled optimization objectives became possible only recently, with the introduction of similarity matching objectives. What explains the success of similarity matching objectives in deriving neural networks with local learning rules? Here, using dimensionality reduction as an example, we introduce several variable substitutions that illuminate the success of similarity matching. We show that the full network objective may be optimized separately for each synapse using local learning rules in both the offline and online settings. We formalize the long-standing intuition of the rivalry between Hebbian and anti-Hebbian rules by formulating a min-max optimization problem. We introduce a novel dimensionality reduction objective using fractional matrix exponents. To illustrate the generality of our approach, we apply it to a novel formulation of dimensionality reduction combined with whitening. We confirm numerically that the networks with learning rules derived from principled objectives perform better than those with heuristic learning rules.

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4. Capturing Spike Variability in Noisy Izhikevich Neurons Using Point Process Generalized Linear Models

Jacob Stærk■Østergaard, Mark Kramer, Uri T. Eden

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To understand neural activity, two broad categories of models exist: statistical and dynamical. While statistical models possess rigorous methods for parameter estimation and goodness-of-fit assessment, dynamical models provide mechanistic insight. In general, these two categories of models are separately applied; understanding the relationships between these modeling approaches remains an area of active research. In this letter, we examine this relationship using simulation. To do so, we first generate spike train data from a well-known dynamical model, the Izhikevich neuron, with a noisy input current. We then fit these spike train data with a statistical model (a generalized linear model, GLM, with multiplicative influences of past spiking). For different levels of noise, we show how the GLM captures both the deterministic features of the Izhikevich neuron and the variability driven by the noise. We conclude that the GLM captures essential features of the simulated spike trains, but for near-deterministic spike trains, goodness-of-fit analyses reveal that the model does not fit very well in a statistical sense; the essential random part of the GLM is not captured.

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5. Mechanism-Based and Input-Output Modeling of the Key Neuronal Connections and Signal Transformations in the CA3-CA1 Regions of the Hippocampus

Kunling Geng, Dae C. Shin, Dong Song, Robert E. Hampson, Samuel A. Deadwyler, Theodore W. Berger, Vasilis Z. Marmarelis

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This letter examines the results of input-output (nonparametric) modeling based on the analysis of data generated by a mechanism-based (parametric) model of CA3-CA1 neuronal connections in the hippocampus. The motivation is to obtain biological insight into the interpretation of such input-output (Volterra-equivalent) models estimated from synthetic data. The insights obtained may be subsequently used to interpretat input-output models extracted from actual experimental data. Specifically, we found that a

simplified parametric model may serve as a useful tool to study the signal transformations in the hippocampal CA3-CA1 regions. Input-output modeling of model-based synthetic data show that GABAergic interneurons are responsible for regulating neuronal excitation, controlling the precision of spike timing, and maintaining network oscillations, in a manner consistent with previous studies. The input-output model obtained from real data exhibits intriguing similarities with its synthetic-data counterpart, demonstrating the importance of a dynamic resonance in the system/model response around 2 Hz to 3 Hz. Using the input-output model from real data as a guide, we may be able to amend the parametric model by incorporating more mechanisms in order to yield better-matching input-output model. The approach we present can also be applied to the study of other neural systems and pathways.

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6. Improved Perceptual Learning by Control of Extracellular GABA Concentration by Astrocytic Gap Junctions

Osamu Hoshino, Meihong Zheng, Kazuo Watanabe

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Learning of sensory cues is believed to rely on synchronous pre- and postsynaptic neuronal firing. Evidence is mounting that such synchronicity is not merely caused by properties of the underlying neuronal network but could also depend on the integrity of gap junctions that connect neurons and astrocytes in networks too. In this perspective, we set out to investigate the effect of astrocytic gap junctions on perceptual learning, introducing a model for coupled neuron-astrocyte networks. In particular, we focus on the fact that astrocytes are rich of GABA transporters (GATs) which can either uptake or release GABA depending on the astrocyte membrane potential, which is a function of local neural activity. We show that GABAergic signaling is a crucial component of intracolumnar neuronal synchronization, thereby promoting learning by neurons in the same cell assembly that are activated by a shared sensory cue. At the same time, we show that this effect can critically depend on astrocytic gap junctions insofar as these latter could synchronize extracellular GABA levels around many neurons and throughout entire cell assemblies. These results are supported by extensive computational arguments and predict that astrocytic gap junctions could improve perceptual learning by controlling extracellular GABA.

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7. Sequence Classification Using Third-Order Moments

Rasmus Troelsgaard, Lars Kai Hansen

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Model-based classification of sequence data using a set of hidden Markov models is a well-known technique. The involved score function, which is often based on the class-conditional likelihood, can, however, be computationally demanding, especially for long data sequences. Inspired by recent theoretical advances in spectral learning of hidden Markov models, we propose a score function based on third-order moments. In particular, we propose to use the Kullback-Leibler divergence between theoretical and empirical third-order moments for classification of sequence data with discrete observations. The proposed method provides lower computational complexity at classification time than the usual likelihood-based methods. In order to demonstrate the properties of the proposed method, we perform classification of both simulated data and empirical data from a human activity recognition study.

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8. Predictive Coding for Dynamic Visual Processing: Development of Functional Hierarchy in a Multiple Spatiotemporal Scales RNN Model

Minkyu Choi, Jun Tani

https://doi.org/10.1162/neco_a_01026 | pp. 237-270

This letter proposes a novel predictive coding type neural network model, the predictive multiple spatiotemporal scales recurrent neural network (P-MSTRNN). The P-MSTRNN learns to predict visually perceived human whole-body cyclic movement patterns by exploiting multiscale spatiotemporal constraints imposed on network dynamics by using differently sized receptive fields as well as different time constant values for each layer. After learning, the network can imitate target movement patterns by inferring or recognizing corresponding intentions by means of the regression of prediction error. Results show that the network can develop a functional hierarchy by developing a different type of dynamic structure at each layer. The letter examines how model performance during pattern generation, as well as predictive imitation, varies depending on the stage of learning. The number of limit cycle attractors corresponding to target movement patterns increases as learning proceeds. Transient dynamics developing early in the learning process successfully perform pattern generation and predictive imitation tasks. The letter concludes that exploitation of transient dynamics facilitates successful task performance during early learning periods.

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9. Temporal Causal Inference with Time Lag

Sizhen Du, Guojie Song, Lei Han, Haikun Hong

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Accurate causal inference among time series helps to better understand the interactive scheme behind the temporal variables. For time series analysis, an unavoidable issue is the existence of time lag among different temporal variables. That is, past evidence would take some time to cause a future effect instead of an immediate response. To model this process, existing approaches commonly adopt a prefixed time window to define the lag. However, in many real-world applications, this parameter may vary among different time series, and it is hard to be predefined with a fixed value. In this letter, we propose to learn the causal relations as well as the lag among different time series simultaneously from data. Specifically, we develop a probabilistic decomposed slab-and-spike (DSS) model to perform the inference by applying a pair of decomposed spike-and-slab variables for the model coefficients, where the first variable is used to estimate the causal relationship and the second one captures the lag information among different temporal variables. For parameter inference, we propose an efficient expectation propagation (EP) algorithm to solve the DSS model. Experimental results conducted on both synthetic and real-world problems demonstrate the effectiveness of the proposed method. The revealed time lag can be well validated by the domain knowledge within the real-world applications.

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10. The Interaction between Semantic Representation and Episodic Memory

Jing Fang, Naima Rüther, Christian Bellebaum, Laurenz Wiskott, Sen Cheng

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The experimental evidence on the interrelation between episodic memory and semantic memory is inconclusive. Are they independent systems, different aspects of a single system, or separate but strongly

interacting systems? Here, we propose a computational role for the interaction between the semantic and episodic systems that might help resolve this debate. We hypothesize that episodic memories are represented as sequences of activation patterns. These patterns are the output of a semantic representational network that compresses the high-dimensional sensory input. We show quantitatively that the accuracy of episodic memory crucially depends on the quality of the semantic representation. We compare two types of semantic representations: appropriate representations, which means that the representation is used to store input sequences that are of the same type as those that it was trained on, and inappropriate representations, which means that stored inputs differ from the training data. Retrieval accuracy is higher for appropriate representations because the encoded sequences are less divergent than those encoded with inappropriate representations. Consistent with our model prediction, we found that human subjects remember some aspects of episodes significantly more accurately if they had previously been familiarized with the objects occurring in the episode, as compared to episodes involving unfamiliar objects. We thus conclude that the interaction with the semantic system plays an important role for episodic memory.

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11. On Rhythms in Neuronal Networks with Recurrent Excitation

Christoph Börgers, Ryusei Melody Takeuchi, Daniel Rosebrock

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We investigate rhythms in networks of neurons with recurrent excitation, that is, with excitatory cells exciting each other. Recurrent excitation can sustain activity even when the cells in the network are driven below threshold, too weak to fire on their own. This sort of "reverberating" activity is often thought to be the basis of working memory. Recurrent excitation can also lead to "runaway" transitions, sudden transitions to high-frequency firing; this may be related to epileptic seizures. Not all fundamental questions about these phenomena have been answered with clarity in the literature. We focus on three questions here: (1) How much recurrent excitation is needed to sustain reverberating activity? How does the answer depend on parameters? (2) Is there a positive minimum frequency of reverberating activity, a positive "onset frequency"? How does it depend on parameters? (3) When do runaway transitions occur? For reduced models, we give mathematical answers to these questions. We also examine computationally to which extent our findings are reflected in the behavior of biophysically more realistic model networks. Our main results can be summarized as follows. (1) Reverberating activity can be fueled by extremely weak slow recurrent excitation, but only by sufficiently strong fast recurrent excitation. (2) The onset of reverberating activity, as recurrent excitation is strengthened or external drive is raised, occurs at a positive frequency. It is faster when the external drive is weaker (and the recurrent excitation stronger). It is slower when the recurrent excitation has a longer decay time constant. (3) Runaway transitions occur only with fast, not with slow, recurrent excitation. We also demonstrate that the relation between reverberating activity fueled by recurrent excitation and runaway transitions can be visualized in an instructive way by a (generalized) cusp catastrophe surface.

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12. Encoding Time in Feedforward Trajectories of a Recurrent Neural Network Model

Nicholas F. Hardy, Dean V. Buonomano

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Brain activity evolves through time, creating trajectories of activity that underlie sensorimotor processing, behavior, and learning and memory. Therefore, understanding the temporal nature of neural dynamics is essential to understanding brain function and behavior. In vivo studies have demonstrated that sequential transient activation of neurons can encode time. However, it remains unclear whether these patterns emerge from feedforward network architectures or from recurrent networks and, furthermore, what role network structure plays in timing. We address these issues using a recurrent neural network (RNN) model with distinct populations of excitatory and inhibitory units. Consistent with experimental data, a single RNN could autonomously produce multiple functionally feedforward trajectories, thus potentially encoding multiple timed motor patterns lasting up to several seconds. Importantly, the model accounted for Weber's law, a hallmark of timing behavior. Analysis of network connectivity revealed that efficiency-a measure of network interconnectedness-decreased as the number of stored trajectories increased. Additionally, the balance of excitation (E) and inhibition (I) shifted toward excitation during each unit's activation time, generating the prediction that observed sequential activity relies on dynamic control of the E/I balance. Our results establish for the first time that the same RNN can generate multiple functionally feedforward patterns of activity as a result of dynamic shifts in the E/I balance imposed by the connectome of the RNN. We conclude that recurrent network architectures account for sequential neural activity, as well as for a fundamental signature of timing behavior: Weber's law.

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13. A Single, Continuously Applied Control Policy for Modeling Reaching Movements with and without Perturbation

Zhe Li, Pietro Mazzoni, Sen Song, Ning Qian

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It has been debated whether kinematic features, such as the number of peaks or decomposed submovements in a velocity profile, indicate the number of discrete motor impulses or result from a continuous control process. The debate is particularly relevant for tasks involving target perturbation, which can alter movement kinematics. To simulate such tasks, finite-horizon models require two preset movement durations to compute two control policies before and after the perturbation. Another model employs infinite- and finite-horizon formulations to determine, respectively, movement durations and control policies, which are updated every time step. We adopted an infinite-horizon optimal feedback control model that, unlike previous approaches, does not preset movement durations or use multiple control policies. It contains both control-dependent and independent noises in system dynamics, state-dependent and independent noises in sensory feedbacks, and different delays and noise levels for visual and proprioceptive feedbacks. We analytically derived an optimal solution that can be applied continuously to move an effector toward a target regardless of whether, when, or where the target jumps. This single policy produces different numbers of peaks and "submovements" in velocity profiles for different conditions and trials. Movements that are slower or perturbed later appear to have more submovements. The model is also consistent with the observation that subjects can perform the perturbation task even without detecting the target jump or seeing their hands during reaching. Finally, because the model incorporates Weber's law via a state representation relative to the target, it explains why initial and terminal visual feedback are, respectively, less and more effective in improving end-point accuracy. Our work suggests that the number of peaks or submovements in a velocity profile does not necessarily reflect the number of motor impulses and that the difference between initial and

terminal feedback does not necessarily imply a transition between open- and closed-loop strategies.

14. A Perceptual-Like Population-Coding Mechanism of Approximate Numerical Averaging

Noam Brezis, Zohar Z. Bronfman, Marius Usher

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Humans possess a remarkable ability to rapidly form coarse estimations of numerical averages. This ability is important for making decisions that are based on streams of numerical or value-based information, as well as for preference formation. Nonetheless, the mechanism underlying rapid approximate numerical averaging remains unknown, and several competing mechanism may account for it. Here, we tested the hypothesis that approximate numerical averaging relies on perceptual-like processes, instantiated by population coding. Participants were presented with rapid sequences of numerical values (four items per second) and were asked to convey the sequence average. We manipulated the sequences' length, variance, and mean magnitude and found that similar to perceptual averaging, the precision of the estimations improves with the length and deteriorates with (higher) variance or (higher) magnitude. To account for the results, we developed a biologically plausible population-coding model and showed that it is mathematically equivalent to a population vector. Using both quantitative and qualitative model comparison methods, we compared the population-coding model to several competing models, such as a step-by-step running average (based on leaky integration) and a midrange model. We found that the data support the population-coding model. We conclude that humans' ability to rapidly form estimations of numerical averages has many properties of the perceptual (intuitive) system rather than the arithmetic, linguistic-based (analytic) system and that population coding is likely to be its underlying mechanism.

15. Statistics of Visual Responses to Image Object Stimuli from Primate AIT Neurons to DNN Neurons

Qiulei Dong, Hong Wang, Zhanyi Hu

https://doi.org/10.1162/neco_a_01039 | pp. 447-476

Under the goal-driven paradigm, Yamins et al. (2014 ; Yamins & DiCarlo, 2016) have shown that by optimizing only the final eight-way categorization performance of a four-layer hierarchical network, not only can its top output layer quantitatively predict IT neuron responses but its penultimate layer can also automatically predict V4 neuron responses. Currently, deep neural networks (DNNs) in the field of computer vision have reached image object categorization performance comparable to that of human beings on ImageNet, a data set that contains 1.3 million training images of 1000 categories. We explore whether the DNN neurons (units in DNNs) possess image object representational statistics similar to monkey IT neurons, particularly when the network becomes deeper and the number of image categories becomes larger, using VGG19, a typical and widely used deep network of 19 layers in the computer vision field. Following Lehky, Kiani, Esteky, and Tanaka (2011 , 2014), where the response statistics of 674 IT neurons to 806 image stimuli are analyzed using three measures (kurtosis, Pareto tail index, and intrinsic dimensionality), we investigate the three issues in this letter using the same three measures: (1) the similarities and differences of the neural response statistics between VGG19 and primate IT cortex, (2) the variation trends of the response statistics of VGG19 neurons at different layers from low to high, and (3) the

variation trends of the response statistics of VGG19 neurons when the numbers of stimuli and neurons increase. We find that the response statistics on both single-neuron selectivity and population sparseness of VGG19 neurons are fundamentally different from those of IT neurons in most cases; by increasing the number of neurons in different layers and the number of stimuli, the response statistics of neurons at different layers from low to high do not substantially change; and the estimated intrinsic dimensionality values at the low convolutional layers of VGG19 are considerably larger than the value of approximately 100 reported for IT neurons in Lehky et al. (2014), whereas those at the high fully connected layers are close to or lower than 100. To the best of our knowledge, this work is the first attempt to analyze the response statistics of DNN neurons with respect to primate IT neurons in image object representation.

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16. Sufficient Dimension Reduction via Direct Estimation of the Gradients of Logarithmic Conditional Densities

Hiroaki Sasaki, Voot Tangkaratt, Gang Niu, Masashi Sugiyama

https://doi.org/10.1162/neco_a_01035 | pp. 477-504

Sufficient dimension reduction (SDR) is aimed at obtaining the low-rank projection matrix in the input space such that information about output data is maximally preserved. Among various approaches to SDR, a promising method is based on the eigendecomposition of the outer product of the gradient of the conditional density of output given input. In this letter, we propose a novel estimator of the gradient of the logarithmic conditional density that directly fits a linear-in-parameter model to the true gradient under the squared loss. Thanks to this simple least-squares formulation, its solution can be computed efficiently in a closed form. Then we develop a new SDR method based on the proposed gradient estimator. We theoretically prove that the proposed gradient estimator, as well as the SDR solution obtained from it, achieves the optimal parametric convergence rate. Finally, we experimentally demonstrate that our SDR method compares favorably with existing approaches in both accuracy and computational efficiency on a variety of artificial and benchmark data sets.

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17. Rank-Optimized Logistic Matrix Regression toward Improved Matrix Data Classification

Jianguang Zhang, Jianmin Jiang

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While existing logistic regression suffers from overfitting and often fails in considering structural information, we propose a novel matrix-based logistic regression to overcome the weakness. In the proposed method, 2D matrices are directly used to learn two groups of parameter vectors along each dimension without vectorization, which allows the proposed method to fully exploit the underlying structural information embedded inside the 2D matrices. Further, we add a joint [Formula: see text]-norm on two parameter matrices, which are organized by aligning each group of parameter vectors in columns. This added co-regularization term has two roles-enhancing the effect of regularization and optimizing the rank during the learning process. With our proposed fast iterative solution, we carried out extensive experiments. The results show that in comparison to both the traditional tensor-based methods and the vector-based regression methods, our proposed solution achieves better performance for matrix data classifications.

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18. Joint Concept Correlation and Feature-Concept Relevance Learning for Multilabel Classification

Xiaowei Zhao, Zhigang Ma, Zhi Li, Zhihui Li

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In recent years, multilabel classification has attracted significant attention in multimedia annotation. However, most of the multilabel classification methods focus only on the inherent correlations existing among multiple labels and concepts and ignore the relevance between features and the target concepts. To obtain more robust multilabel classification results, we propose a new multilabel classification method aiming to capture the correlations among multiple concepts by leveraging hypergraph that is proved to be beneficial for relational learning. Moreover, we consider mining feature-concept relevance, which is often overlooked by many multilabel learning algorithms. To better show the feature-concept relevance, we impose a sparsity constraint on the proposed method. We compare the proposed method with several other multilabel classification methods and evaluate the classification performance by mean average precision on several data sets. The experimental results show that the proposed method outperforms the state-of-the-art methods.

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19. Feedforward Approximations to Dynamic Recurrent Network Architectures

Dylan R. Muir

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Recurrent neural network architectures can have useful computational properties, with complex temporal dynamics and input-sensitive attractor states. However, evaluation of recurrent dynamic architectures requires solving systems of differential equations, and the number of evaluations required to determine their response to a given input can vary with the input or can be indeterminate altogether in the case of oscillations or instability. In feedforward networks, by contrast, only a single pass through the network is needed to determine the response to a given input. Modern machine learning systems are designed to operate efficiently on feedforward architectures. We hypothesized that two-layer feedforward architectures with simple, deterministic dynamics could approximate the responses of single-layer recurrent network architectures. By identifying the fixed-point responses of a given recurrent network, we trained two-layer networks to directly approximate the fixed-point response to a given input. These feedforward networks then embodied useful computations, including competitive interactions, information transformations, and noise rejection. Our approach was able to find useful approximations to recurrent networks, which can then be evaluated in linear and deterministic time complexity.

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20. STDP-Compatible Approximation of Backpropagation in an Energy-Based Model

Yoshua Bengio, Thomas Mésnard, Asja Fischer, Saizheng Zhang, Yuhuai Wu

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We show that Langevin Markov chain Monte Carlo inference in an energy-based model with latent variables has the property that the early steps of inference, starting from a stationary point, correspond to propagating

error gradients into internal layers, similar to backpropagation. The backpropagated error is with respect to output units that have received an outside driving force pushing them away from the stationary point. Backpropagated error gradients correspond to temporal derivatives with respect to the activation of hidden units. These lead to a weight update proportional to the product of the presynaptic firing rate and the temporal rate of change of the postsynaptic firing rate. Simulations and a theoretical argument suggest that this rate-based update rule is consistent with those associated with spike-timing-dependent plasticity. The ideas presented in this article could be an element of a theory for explaining how brains perform credit assignment in deep hierarchies as efficiently as backpropagation does, with neural computation corresponding to both approximate inference in continuous-valued latent variables and error backpropagation, at the same time.

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21. Improving Spiking Dynamical Networks: Accurate Delays, Higher-Order Synapses, and Time Cells

Aaron R. Voelker, Chris Eliasmith

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Researchers building spiking neural networks face the challenge of improving the biological plausibility of their model networks while maintaining the ability to quantitatively characterize network behavior. In this work, we extend the theory behind the neural engineering framework (NEF), a method of building spiking dynamical networks, to permit the use of a broad class of synapse models while maintaining prescribed dynamics up to a given order. This theory improves our understanding of how low-level synaptic properties alter the accuracy of high-level computations in spiking dynamical networks. For completeness, we provide characterizations for both continuous-time (i.e., analog) and discrete-time (i.e., digital) simulations. We demonstrate the utility of these extensions by mapping an optimal delay line onto various spiking dynamical networks using higher-order models of the synapse. We show that these networks nonlinearly encode rolling windows of input history, using a scale invariant representation, with accuracy depending on the frequency content of the input signal. Finally, we reveal that these methods provide a novel explanation of time cell responses during a delay task, which have been observed throughout hippocampus, striatum, and cortex.

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22. Deep Learning with Dynamic Spiking Neurons and Fixed Feedback Weights

Arash Samadi, Timothy Lillicrap, Douglas Tweed

https://doi.org/10.1162/neco_a_00929 | pp. 578-602

Recent work in computer science has shown the power of deep learning driven by the backpropagation algorithm in networks of artificial neurons. But real neurons in the brain are different from most of these artificial ones in at least three crucial ways: they emit spikes rather than graded outputs, their inputs and outputs are related dynamically rather than by piecewise-smooth functions, and they have no known way to coordinate arrays of synapses in separate forward and feedback pathways so that they change simultaneously and identically, as they do in backpropagation. Given these differences, it is unlikely that current deep learning algorithms can operate in the brain, but we that show these problems can be solved by two simple devices: learning rules can approximate dynamic input-output relations with piecewise-smooth functions, and a variation on the feedback alignment algorithm can train deep networks without having to

coordinate forward and feedback synapses. Our results also show that deep spiking networks learn much better if each neuron computes an intracellular teaching signal that reflects that cell's nonlinearity. With this mechanism, networks of spiking neurons show useful learning in synapses at least nine layers upstream from the output cells and perform well compared to other spiking networks in the literature on the MNIST digit recognition task.

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23. Interpretation of the Precision Matrix and Its Application in Estimating Sparse Brain Connectivity during Sleep Spindles from Human Electrocorticography Recordings

Anup Das, Aaron L. Sampson, Claudia Lainscsek, Lyle Muller, Wutu Lin, John C. Doyle, Sydney S. Cash, Eric Halgren, Terrence J. Sejnowski

https://doi.org/10.1162/neco_a_00936 | pp. 603-642

The correlation method from brain imaging has been used to estimate functional connectivity in the human brain. However, brain regions might show very high correlation even when the two regions are not directly connected due to the strong interaction of the two regions with common input from a third region. One previously proposed solution to this problem is to use a sparse regularized inverse covariance matrix or precision matrix (SRPM) assuming that the connectivity structure is sparse. This method yields partial correlations to measure strong direct interactions between pairs of regions while simultaneously removing the influence of the rest of the regions, thus identifying regions that are conditionally independent. To test our methods, we first demonstrated conditions under which the SRPM method could indeed find the true physical connection between a pair of nodes for a spring-mass example and an RC circuit example. The recovery of the connectivity structure using the SRPM method can be explained by energy models using the Boltzmann distribution. We then demonstrated the application of the SRPM method for estimating brain connectivity during stage 2 sleep spindles from human electrocorticography (ECoG) recordings using an [Formula: see text] electrode array. The ECoG recordings that we analyzed were from a 32-year-old male patient with long-standing pharmaco-resistant left temporal lobe complex partial epilepsy. Sleep spindles were automatically detected using delay differential analysis and then analyzed with SRPM and the Louvain method for community detection. We found spatially localized brain networks within and between neighboring cortical areas during spindles, in contrast to the case when sleep spindles were not present.

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24. Learning Midlevel Auditory Codes from Natural Sound Statistics

Wiktor M■ynarski, Josh H. McDermott

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Interaction with the world requires an organism to transform sensory signals into representations in which behaviorally meaningful properties of the environment are made explicit. These representations are derived through cascades of neuronal processing stages in which neurons at each stage recode the output of preceding stages. Explanations of sensory coding may thus involve understanding how low-level patterns are combined into more complex structures. To gain insight into such midlevel representations for sound, we designed a hierarchical generative model of natural sounds that learns combinations of spectrotemporal features from natural stimulus statistics. In the first layer, the model forms a sparse convolutional code of spectrograms using a dictionary of learned spectrotemporal kernels. To generalize from specific kernel

activation patterns, the second layer encodes patterns of time-varying magnitude of multiple first-layer coefficients. When trained on corpora of speech and environmental sounds, some second-layer units learned to group similar spectrotemporal features. Others instantiate opponency between distinct sets of features. Such groupings might be instantiated by neurons in the auditory cortex, providing a hypothesis for midlevel neuronal computation.

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25. Effects of Small-World Rewiring Probability and Noisy Synaptic Conductivity on Slow Waves: Cortical Network

Ramazan Teki■n, M. Emin Ta■luk

https://doi.org/10.1162/neco_a_00932 | pp. 679-715

Physiological rhythms play a critical role in the functional development of living beings. Many biological functions are executed with an interaction of rhythms produced by internal characteristics of scores of cells. While synchronized oscillations may be associated with normal brain functions, anomalies in these oscillations may cause or relate the emergence of some neurological or neuropsychological pathologies. This study was designed to investigate the effects of topological structure and synaptic conductivity noise on the spatial synchronization and temporal rhythmicity of the waves generated by cells in the network. Because of holding the ability of clustering and randomizing with change of parameters, small-world (SW) network topology was chosen. The oscillatory activity of network was tried out by manipulating an insulated SW, cortical network model whose morphology is very close to real world. According to the obtained results, it was observed that at the optimal probabilistic rates of conductivity noise and rewiring of SW, powerful synchronized oscillatory small waves are generated in relation to the internal dynamics of cells, which are in line with the network's input. These two parameters were observed to be quite effective on the excitation-inhibition balance of the network. Accordingly, it may be suggested that the topological dynamics of SW and noisy synaptic conductivity may be associated with the normal and abnormal development of neurobiological structure.

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26. Spiking Neural Classifier with Lumped Dendritic Nonlinearity and Binary Synapses: A Current Mode VLSI Implementation and Analysis

Aritra Bhaduri, Amitava Banerjee, Subhrajit Roy, Sougata Kumar Kar, Arindam Basu

https://doi.org/10.1162/neco_a_01045 | pp. 723-760

We present a neuromorphic current mode implementation of a spiking neural classifier with lumped square law dendritic nonlinearity. It has been shown previously in software simulations that such a system with binary synapses can be trained with structural plasticity algorithms to achieve comparable classification accuracy with fewer synaptic resources than conventional algorithms. We show that even in real analog systems with manufacturing imperfections (CV of 23.5% and 14.4% for dendritic branch gains and leaks respectively), this network is able to produce comparable results with fewer synaptic resources. The chip fabricated in [Formula: see text]m complementary metal oxide semiconductor has eight dendrites per cell and uses two opposing cells per class to cancel common-mode inputs. The chip can operate down to a [Formula: see text] V and dissipates 19 nW of static power per neuronal cell and [Formula: see text] 125 pJ/spike. For two-class classification problems of high-dimensional rate encoded binary patterns, the hardware achieves comparable performance as software implementation of the same with only about a

0.5% reduction in accuracy. On two UCI data sets, the IC integrated circuit has classification accuracy comparable to standard machine learners like support vector machines and extreme learning machines while using two to five times binary synapses. We also show that the system can operate on mean rate encoded spike patterns, as well as short bursts of spikes. To the best of our knowledge, this is the first attempt in hardware to perform classification exploiting dendritic properties and binary synapses.

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27. Multisensory Bayesian Inference Depends on Synapse Maturation during Training: Theoretical Analysis and Neural Modeling Implementation

Mauro Ursino, Cristiano Cuppini, Elisa Magosso

https://doi.org/10.1162/neco_a_00935 | pp. 735-782

Recent theoretical and experimental studies suggest that in multisensory conditions, the brain performs a near-optimal Bayesian estimate of external events, giving more weight to the more reliable stimuli. However, the neural mechanisms responsible for this behavior, and its progressive maturation in a multisensory environment, are still insufficiently understood. The aim of this letter is to analyze this problem with a neural network model of audiovisual integration, based on probabilistic population coding-the idea that a population of neurons can encode probability functions to perform Bayesian inference. The model consists of two chains of unisensory neurons (auditory and visual) topologically organized. They receive the corresponding input through a plastic receptive field and reciprocally exchange plastic cross-modal synapses, which encode the spatial co-occurrence of visual-auditory inputs. A third chain of multisensory neurons performs a simple sum of auditory and visual excitations. The work includes a theoretical part and a computer simulation study. We show how a simple rule for synapse learning (consisting of Hebbian reinforcement and a decay term) can be used during training to shrink the receptive fields and encode the unisensory likelihood functions. Hence, after training, each unisensory area realizes a maximum likelihood estimate of stimulus position (auditory or visual). In cross-modal conditions, the same learning rule can encode information on prior probability into the cross-modal synapses. Computer simulations confirm the theoretical results and show that the proposed network can realize a maximum likelihood estimate of auditory (or visual) positions in unimodal conditions and a Bayesian estimate, with moderate deviations from optimality, in cross-modal conditions. Furthermore, the model explains the ventriloquism illusion and, looking at the activity in the multimodal neurons, explains the automatic reweighting of auditory and visual inputs on a trial-by-trial basis, according to the reliability of the individual cues.

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28. From Winner-Takes-All to Winners-Share-All: Exploiting the Information Capacity in Temporal Codes

Melika Payvand, Luke Theogarajan

https://doi.org/10.1162/neco_a_01047 | pp. 761-791

In this letter, we have implemented and compared two neural coding algorithms in the networks of spiking neurons: Winner-takes-all (WTA) and winners-share-all (WSA). Winners-Share-All exploits the code space provided by the temporal code by training a different combination of [Formula: see text] out of [Formula: see text] neurons to fire together in response to different patterns, while WTA uses a one-hot-coding to respond to distinguished patterns. Using WSA, the maximum value of [Formula: see text] in order to maximize information capacity using [Formula: see text] output neurons was theoretically determined and utilized. A

small proof-of-concept classification problem was applied to a spiking neural network using both algorithms to classify 14 letters of English alphabet with an image size of 15 [Formula: see text] 15 pixels. For both schemes, a modified spike-timing-dependent-plasticity (STDP) learning rule has been used to train the spiking neurons in an unsupervised fashion. The performance and the number of neurons required to perform this computation are compared between the two algorithms. We show that by tolerating a small drop in performance accuracy (84% in WSA versus 91% in WTA), we are able to reduce the number of output neurons by more than a factor of two. We show how the reduction in the number of neurons will increase as the number of patterns increases. The reduction in the number of output neurons would then proportionally reduce the number of training parameters, which requires less memory and hence speeds up the computation, and in the case of neuromorphic implementation on silicon, would take up much less area.

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29. Spike-Centered Jitter Can Mistake Temporal Structure

Jonathan Platkiewicz, Eran Stark, Asohan Amarasingham

https://doi.org/10.1162/neco_a_00927 | pp. 783-803

Jitter-type spike resampling methods are routinely applied in neurophysiology for detecting temporal structure in spike trains (point processes). Several variations have been proposed. The concern has been raised, based on numerical experiments involving Poisson spike processes, that such procedures can be conservative. We study the issue and find it can be resolved by reemphasizing the distinction between spike-centered (basic) jitter and interval jitter. Focusing on spiking processes with no temporal structure, interval jitter generates an exact hypothesis test, guaranteeing valid conclusions. In contrast, such a guarantee is not available for spike-centered jitter. We construct explicit examples in which spike-centered jitter hallucinates temporal structure, in the sense of exaggerated false-positive rates. Finally, we illustrate numerically that Poisson approximations to jitter computations, while computationally efficient, can also result in inaccurate hypothesis tests. We highlight the value of classical statistical frameworks for guiding the design and interpretation of spike resampling methods.

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30. Quantifying Strength of Chaos in the Population Firing Rate of Neurons

Takashi Kanamaru

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In this study, I considered quantifying the strength of chaos in the population firing rate of a pulse-coupled neural network. In particular, I considered the dynamics where the population firing rate is chaotic and the firing of each neuron is stochastic. I calculated a time histogram of firings to show the variation in the population firing rate over time. To smooth this histogram, I used Bayesian adaptive regression splines and a gaussian filter. The nonlinear prediction method, based on reconstruction, was applied to a sequence of interpeak intervals in the smoothed time histogram of firings. I propose the use of the sum of nonlinearity as a quantifier of the strength of chaos. When applying this method to the firings of a pulse-coupled neural network, the sum of nonlinearity was seen to satisfy three properties for quantifying the strength of chaos. First, it can be calculated from spiking data alone. Second, it takes large values when applied to firings that are confirmed, theoretically or numerically, to be chaotic. Third, it reflects the strength of chaos of the original dynamics.

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31. Analysis of Online Composite Mirror Descent Algorithm

Yunwen Lei, Ding■Xuan Zhou

https://doi.org/10.1162/neco_a_00930 | pp. 825-860

We study the convergence of the online composite mirror descent algorithm, which involves a mirror map to reflect the geometry of the data and a convex objective function consisting of a loss and a regularizer possibly inducing sparsity. Our error analysis provides convergence rates in terms of properties of the strongly convex differentiable mirror map and the objective function. For a class of objective functions with Hölder continuous gradients, the convergence rates of the excess (regularized) risk under polynomially decaying step sizes have the order [Formula: see text] after [Formula: see text] iterates. Our results improve the existing error analysis for the online composite mirror descent algorithm by avoiding averaging and removing boundedness assumptions, and they sharpen the existing convergence rates of the last iterate for online gradient descent without any boundedness assumptions. Our methodology mainly depends on a novel error decomposition in terms of an excess Bregman distance, refined analysis of self-bounding properties of the objective function, and the resulting one-step progress bounds.

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32. Solving Nonlinearly Separable Classifications in a Single-Layer Neural Network

Nolan Conaway, Kenneth J. Kurtz

https://doi.org/10.1162/neco_a_00931 | pp. 861-866

Since the work of Minsky and Papert (1969), it has been understood that single-layer neural networks cannot solve nonlinearly separable classifications (i.e., XOR). We describe and test a novel divergent autoassociative architecture capable of solving nonlinearly separable classifications with a single layer of weights. The proposed network consists of class-specific linear autoassociators. The power of the model comes from treating classification problems as within-class feature prediction rather than directly optimizing a discriminant function. We show unprecedented learning capabilities for a simple, single-layer network (i.e., solving XOR) and demonstrate that the famous limitation in acquiring nonlinearly separable problems is not just about the need for a hidden layer; it is about the choice between directly predicting classes or learning to classify indirectly by predicting features.

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33. Unifying Adversarial Training Algorithms with Data Gradient Regularization

Alexander G. Ororbia, Daniel Kifer, C. Lee Giles

https://doi.org/10.1162/neco_a_00928 | pp. 867-887

Many previous proposals for adversarial training of deep neural nets have included directly modifying the gradient, training on a mix of original and adversarial examples, using contractive penalties, and approximately optimizing constrained adversarial objective functions. In this article, we show that these proposals are actually all instances of optimizing a general, regularized objective we call DataGrad. Our proposed DataGrad framework, which can be viewed as a deep extension of the layerwise contractive autoencoder penalty, cleanly simplifies prior work and easily allows extensions such as adversarial training with multitask cues. In our experiments, we find that the deep gradient regularization of DataGrad (which

also has L1 and L2 flavors of regularization) outperforms alternative forms of regularization, including classical L1, L2, and multitask, on both the original data set and adversarial sets. Furthermore, we find that combining multitask optimization with DataGrad adversarial training results in the most robust performance.

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34. Information Maximization Explains the Sparseness of Presynaptic Neural Response

Minjoon Kouh

https://doi.org/10.1162/neco_a_00943 | pp. 888-896

In a sensory neural network, where a population of presynaptic neurons sends information to a downstream neuron, maximizing information transmission depends on utilizing the full operating range of the output of the postsynaptic neuron. Because the convergence of presynaptic inputs naturally biases higher outputs, a sparse input distribution would counter such bias and optimize information transmission.

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35. On the Dynamical Interplay of Positive and Negative Affects

Jonathan Touboul, Alberto Romagnoni, Robert M. Schwartz

https://doi.org/10.1162/neco_a_00942 | pp. 897-936

Emotional disorders and psychological flourishing are the result of complex interactions between positive and negative affects that depend on external events and the subject's internal representations. Based on psychological data, we mathematically model the dynamical balance between positive and negative affects as a function of the response to external positive and negative events. This modeling allows the investigation of the relative impact of two leading forms of therapy on affect balance. The model uses a delay differential equation to analytically study the bifurcation diagram of the system. We compare the results of the model to psychological data on a single, recurrently depressed patient who was administered the two types of therapies considered (coping focused versus affect focused). The model leads to the prediction that stabilization at a normal state may rely on evaluating one's emotional state through a historical ongoing emotional state rather than in a narrow present window. The simple mathematical model proposed here offers a theoretical framework for investigating the temporal process of change and parameters of resilience to relapse.

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36. Evolving Network Model That Almost Regenerates Epileptic Data

G. Manjunath

https://doi.org/10.1162/neco_a_00941 | pp. 937-967

In many realistic networks, the edges representing the interactions between nodes are time varying. Evidence is growing that the complex network that models the dynamics of the human brain has time-varying interconnections, that is, the network is evolving. Based on this evidence, we construct a patient- and data-specific evolving network model (comprising discrete-time dynamical systems) in which epileptic seizures or their terminations in the brain are also determined by the nature of the time-varying interconnections between the nodes. A novel and unique feature of our methodology is that the evolving network model remembers the data from which it was conceived from, in the sense that it evolves to almost

regenerate the patient data even on presenting an arbitrary initial condition to it. We illustrate a potential utility of our methodology by constructing an evolving network from clinical data that aids in identifying an approximate seizure focus; nodes in such a theoretically determined seizure focus are outgoing hubs that apparently act as spreaders of seizures. We also point out the efficacy of removal of such spreaders in limiting seizures.

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37. Multiway Array Decomposition of EEG Spectrum: Implications of Its Stability for the Exploration of Large-Scale Brain Networks

Radek Mare■ek, Martin Lamoš, René Labounek, Marek Barto■, T. Slavicek, Michal Mikl, Ivan Rektor, Milan Brázdil

https://doi.org/10.1162/neco_a_00933 | pp. 968-989

Multiway array decomposition methods have been shown to be promising statistical tools for identifying neural activity in the EEG spectrum. They blindly decompose the EEG spectrum into spatial-temporal-spectral patterns by taking into account inherent relationships among signals acquired at different frequencies and sensors. Our study evaluates the stability of spatial-temporal-spectral patterns derived by one particular method, parallel factor analysis (PARAFAC). We focused on patterns' stability over time and in population and divided the complete data set containing data from 50 healthy subjects into several subsets. Our results suggest that the patterns are highly stable in time, as well as among different subgroups of subjects. Further, we show with simultaneously acquired fMRI data that power fluctuations of some patterns have stable correspondence to hemodynamic fluctuations in large-scale brain networks. We did not find such correspondence for power fluctuations in standard frequency bands, the common way of dealing with EEG data. Altogether, our results suggest that PARAFAC is a suitable method for research in the field of large-scale brain networks and their manifestation in EEG signal.

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38. Maximum Pseudolikelihood Estimation for Model-Based Clustering of Time Series Data

Hien D. Nguyen, Geoffrey J. McLachlan, Pierre Orban, Pierre Bellec, Andrew L. Janke

https://doi.org/10.1162/neco_a_00938 | pp. 990-1020

Mixture of autoregressions (MoAR) models provide a model-based approach to the clustering of time series data. The maximum likelihood (ML) estimation of MoAR models requires evaluating products of large numbers of densities of normal random variables. In practical scenarios, these products converge to zero as the length of the time series increases, and thus the ML estimation of MoAR models becomes infeasible without the use of numerical tricks. We propose a maximum pseudolikelihood (MPL) estimation approach as an alternative to the use of numerical tricks. The MPL estimator is proved to be consistent and can be computed with an EM (expectation-maximization) algorithm. Simulations are used to assess the performance of the MPL estimator against that of the ML estimator in cases where the latter was able to be calculated. An application to the clustering of time series data arising from a resting state fMRI experiment is presented as a demonstration of the methodology.

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39. An in Silico, Biomarker-Based Method for the Evaluation of Virtual Neuropsychiatric Drug Effects

Peter J. Siekmeier

https://doi.org/10.1162/neco_a_00944 | pp. 1021-1052

The recent explosion in neuroscience research has markedly increased our understanding of the neurobiological correlates of many psychiatric illnesses, but this has unfortunately not translated into more effective pharmacologic treatments for these conditions. At the same time, researchers have increasingly sought out biological markers, or biomarkers, as a way to categorize psychiatric illness, as these are felt to be closer to underlying genetic and neurobiological vulnerabilities. While biomarker-based drug discovery approaches have tended to employ in vivo (e.g., rodent) or in vitro test systems, relatively little attention has been paid to the potential of computational, or in silico, methodologies. Here we describe such a methodology, using as an example a biophysically detailed computational model of hippocampus that is made to generate putative schizophrenia biomarkers by the inclusion of a number of neuropathological changes that have been associated with the illness (NMDA system deficit, decreased neural connectivity, hyperdopaminergia). We use the specific inability to attune to gamma band (40 Hz) auditory stimulus as our illness biomarker. We expose this system to a large number of virtual medications, defined by systematic variation of model parameters corresponding to five cellular-level effects. The potential efficacy of virtual medications is determined by a wellness metric (WM) that we have developed. We identify a number of virtual agents that consist of combinations of mechanisms, which are not simply reversals of the causative lesions. The manner in which this methodology could be extended to other neuropsychiatric conditions, such as Alzheimer's disease, autism, and fragile X syndrome, is discussed.

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40. Semisupervised, Multilabel, Multi-Instance Learning for Structured Data

Hossein Soleimani, David J. Miller

https://doi.org/10.1162/neco_a_00939 | pp. 1053-1102

Many classification tasks require both labeling objects and determining label associations for parts of each object. Example applications include labeling segments of images or determining relevant parts of a text document when the training labels are available only at the image or document level. This task is usually referred to as multi-instance (MI) learning, where the learner typically receives a collection of labeled (or sometimes unlabeled) bags, each containing several segments (instances). We propose a semisupervised MI learning method for multilabel classification. Most MI learning methods treat instances in each bag as independent and identically distributed samples. However, in many practical applications, instances are related to each other and should not be considered independent. Our model discovers a latent low-dimensional space that captures structure within each bag. Further, unlike many other MI learning methods, which are primarily developed for binary classification, we model multiple classes jointly, thus also capturing possible dependencies between different classes. We develop our model within a semisupervised framework, which leverages both labeled and, typically, a larger set of unlabeled bags for training. We develop several efficient inference methods for our model. We first introduce a Markov chain Monte Carlo method for inference, which can handle arbitrary relations between bag labels and instance labels, including the standard hard-max MI assumption. We also develop an extension of our model that uses stochastic variational Bayes methods for inference, and thus scales better to massive data sets. Experiments show that our approach outperforms several MI learning and standard classification methods on both bag-level and

instance-level label prediction. All code for replicating our experiments is available from https://github.com/hsoleimani/MLTM .

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41. Parameter Estimation of Nonlinear Systems by Dynamic Cuckoo Search

Qixiang Liao, Shudao Zhou, Hanqing Shi, Weilai Shi

https://doi.org/10.1162/neco_a_00946 | pp. 1103-1123

In order to address with the problem of the traditional or improved cuckoo search (CS) algorithm, we propose a dynamic adaptive cuckoo search with crossover operator (DACS-CO) algorithm. Normally, the parameters of the CS algorithm are kept constant or adapted by empirical equation that may result in decreasing the efficiency of the algorithm. In order to solve the problem, a feedback control scheme of algorithm parameters is adopted in cuckoo search; Rechenberg's 1/5 criterion, combined with a learning strategy, is used to evaluate the evolution process. In addition, there are no information exchanges between individuals for cuckoo search algorithm. To promote the search progress and overcome premature convergence, the multiple-point random crossover operator is merged into the CS algorithm to exchange information between individuals and improve the diversification and intensification of the population. The performance of the proposed hybrid algorithm is investigated through different nonlinear systems, with the numerical results demonstrating that the method can estimate parameters accurately and efficiently. Finally, we compare the results with the standard CS algorithm, orthogonal learning cuckoo search algorithm (OLCS), an adaptive and simulated annealing operation with the cuckoo search algorithm (ACS-SA), a genetic algorithm (GA), a particle swarm optimization algorithm (PSO), and a genetic simulated annealing algorithm (GA-SA). Our simulation results demonstrate the effectiveness and superior performance of the proposed algorithm.

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42. Avoiding Optimal Mean ■_{2,1}-Norm Maximization-Based Robust PCA for Reconstruction

Minnan Luo, Feiping Nie, Xiaojun Chang, Yi Yang, Alexander G. Hauptmann, Qinghua Zheng

https://doi.org/10.1162/neco_a_00937 | pp. 1124-1150

Robust principal component analysis (PCA) is one of the most important dimension-reduction techniques for handling high-dimensional data with outliers. However, most of the existing robust PCA presupposes that the mean of the data is zero and incorrectly utilizes the average of data as the optimal mean of robust PCA. In fact, this assumption holds only for the squared [Formula: see text]-norm-based traditional PCA. In this letter, we equivalently reformulate the objective of conventional PCA and learn the optimal projection directions by maximizing the sum of projected difference between each pair of instances based on [Formula: see text]-norm. The proposed method is robust to outliers and also invariant to rotation. More important, the reformulated objective not only automatically avoids the calculation of optimal mean and makes the assumption of centered data unnecessary, but also theoretically connects to the minimization of reconstruction error. To solve the proposed nonsmooth problem, we exploit an efficient optimization algorithm to soften the contributions from outliers by reweighting each data point iteratively. We theoretically analyze the convergence and computational complexity of the proposed algorithm. Extensive experimental results on several benchmark data sets illustrate the effectiveness and superiority of the proposed method.

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43. Parameter Identifiability in Statistical Machine Learning: A Review

Zhi-Yong Ran, Bao-Gang Hu

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This review examines the relevance of parameter identifiability for statistical models used in machine learning. In addition to defining main concepts, we address several issues of identifiability closely related to machine learning, showing the advantages and disadvantages of state-of-the-art research and demonstrating recent progress. First, we review criteria for determining the parameter structure of models from the literature. This has three related issues: parameter identifiability, parameter redundancy, and reparameterization. Second, we review the deep influence of identifiability on various aspects of machine learning from theoretical and application viewpoints. In addition to illustrating the utility and influence of identifiability, we emphasize the interplay among identifiability theory, machine learning, mathematical statistics, information theory, optimization theory, information geometry, Riemann geometry, symbolic computation, Bayesian inference, algebraic geometry, and others. Finally, we present a new perspective together with the associated challenges.

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44. Using Inspiration from Synaptic Plasticity Rules to Optimize Traffic Flow in Distributed Engineered Networks

Jonathan Y. Suen, Saket Navlakha

https://doi.org/10.1162/neco_a_00945 | pp. 1204-1228

Controlling the flow and routing of data is a fundamental problem in many distributed networks, including transportation systems, integrated circuits, and the Internet. In the brain, synaptic plasticity rules have been discovered that regulate network activity in response to environmental inputs, which enable circuits to be stable yet flexible. Here, we develop a new neuro-inspired model for network flow control that depends only on modifying edge weights in an activity-dependent manner. We show how two fundamental plasticity rules, long-term potentiation and long-term depression, can be cast as a distributed gradient descent algorithm for regulating traffic flow in engineered networks. We then characterize, both by simulation and analytically, how different forms of edge-weight-update rules affect network routing efficiency and robustness. We find a close correspondence between certain classes of synaptic weight update rules derived experimentally in the brain and rules commonly used in engineering, suggesting common principles to both.

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45. An Approximation of the Error Backpropagation Algorithm in a Predictive Coding Network with Local Hebbian Synaptic Plasticity

James C. R. Whittington, Rafa■ Bogacz

https://doi.org/10.1162/neco_a_00949 | pp. 1229-1262

To efficiently learn from feedback, cortical networks need to update synaptic weights on multiple levels of cortical hierarchy. An effective and well-known algorithm for computing such changes in synaptic weights is the error backpropagation algorithm. However, in this algorithm, the change in synaptic weights is a complex function of weights and activities of neurons not directly connected with the synapse being modified, whereas the changes in biological synapses are determined only by the activity of presynaptic and postsynaptic neurons. Several models have been proposed that approximate the backpropagation algorithm

with local synaptic plasticity, but these models require complex external control over the network or relatively complex plasticity rules. Here we show that a network developed in the predictive coding framework can efficiently perform supervised learning fully autonomously, employing only simple local Hebbian plasticity. Furthermore, for certain parameters, the weight change in the predictive coding model converges to that of the backpropagation algorithm. This suggests that it is possible for cortical networks with simple Hebbian synaptic plasticity to implement efficient learning algorithms in which synapses in areas on multiple levels of hierarchy are modified to minimize the error on the output.

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46. Modulation of Context-Dependent Spatiotemporal Patterns within Packets of Spiking Activity

Miho Itoh, Timothée Leleu

https://doi.org/10.1162/neco_a_00952 | pp. 1263-1292

Recent experiments have shown that stereotypical spatiotemporal patterns occur during brief packets of spiking activity in the cortex, and it has been suggested that top-down inputs can modulate these patterns according to the context. We propose a simple model that may explain important features of these experimental observations and is analytically tractable. The key mechanism underlying this model is that context-dependent top-down inputs can modulate the effective connection strengths between neurons because of short-term synaptic depression. As a result, the degree of synchrony and, in turn, the spatiotemporal patterns of spiking activity that occur during packets are modulated by the top-down inputs. This is shown using an analytical framework, based on avalanche dynamics, that allows calculating the probability that a given neuron spikes during a packet and numerical simulations. Finally, we show that the spatiotemporal patterns that replay previously experienced sequential stimuli and their binding with their corresponding context can be learned because of spike-timing-dependent plasticity.

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47. Variational Latent Gaussian Process for Recovering Single-Trial Dynamics from Population Spike Trains

Yuan Zhao, Il Memming Park

https://doi.org/10.1162/neco_a_00953 | pp. 1293-1316

When governed by underlying low-dimensional dynamics, the interdependence of simultaneously recorded populations of neurons can be explained by a small number of shared factors, or a low-dimensional trajectory. Recovering these latent trajectories, particularly from single-trial population recordings, may help us understand the dynamics that drive neural computation. However, due to the biophysical constraints and noise in the spike trains, inferring trajectories from data is a challenging statistical problem in general. Here, we propose a practical and efficient inference method, the variational latent gaussian process (vLGP). The vLGP combines a generative model with a history-dependent point process observation, together with a smoothness prior on the latent trajectories. The vLGP improves on earlier methods for recovering latent trajectories, which assume either observation models inappropriate for point processes or linear dynamics. We compare and validate vLGP on both simulated data sets and population recordings from the primary visual cortex. In the V1 data set, we find that vLGP achieves substantially higher performance than previous methods for predicting omitted spike trains, as well as capturing both the toroidal topology of visual stimuli space and the noise correlation. These results show that vLGP is a robust method with the potential to

reveal hidden neural dynamics from large-scale neural recordings.

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48. Fast Estimation of Approximate Matrix Ranks Using Spectral Densities

Shashanka Ubaru, Yousef Saad, Abd■Krim Seghouane

https://doi.org/10.1162/neco_a_00951 | pp. 1317-1351

Many machine learning and data-related applications require the knowledge of approximate ranks of large data matrices at hand. This letter presents two computationally inexpensive techniques to estimate the approximate ranks of such matrices. These techniques exploit approximate spectral densities, popular in physics, which are probability density distributions that measure the likelihood of finding eigenvalues of the matrix at a given point on the real line. Integrating the spectral density over an interval gives the eigenvalue count of the matrix in that interval. Therefore, the rank can be approximated by integrating the spectral density over a carefully selected interval. Two different approaches are discussed to estimate the approximate rank, one based on Chebyshev polynomials and the other based on the Lanczos algorithm. In order to obtain the appropriate interval, it is necessary to locate a gap between the eigenvalues that correspond to noise and the relevant eigenvalues that contribute to the matrix rank. A method for locating this gap and selecting the interval of integration is proposed based on the plot of the spectral density. Numerical experiments illustrate the performance of these techniques on matrices from typical applications.

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49. Unsupervised 2D Dimensionality Reduction with Adaptive Structure Learning

Xiaowei Zhao, Feiping Nie, Sen Wang, Jun Guo, Pengfei Xu, Xiaojiang Chen

https://doi.org/10.1162/neco_a_00950 | pp. 1352-1374

In recent years, unsupervised two-dimensional (2D) dimensionality reduction methods for unlabeled large-scale data have made progress. However, performance of these degrades when the learning of similarity matrix is at the beginning of the dimensionality reduction process. A similarity matrix is used to reveal the underlying geometry structure of data in unsupervised dimensionality reduction methods. Because of noise data, it is difficult to learn the optimal similarity matrix. In this letter, we propose a new dimensionality reduction model for 2D image matrices: unsupervised 2D dimensionality reduction with adaptive structure learning (DRASL). Instead of using a predetermined similarity matrix to characterize the underlying geometry structure of the original 2D image space, our proposed approach involves the learning of a similarity matrix in the procedure of dimensionality reduction. To realize a desirable neighbors assignment after dimensionality reduction, we add a constraint to our model such that there are exact [Formula: see text] connected components in the final subspace. To accomplish these goals, we propose a unified objective function to integrate dimensionality reduction, the learning of the similarity matrix, and the adaptive learning of neighbors assignment into it. An iterative optimization algorithm is proposed to solve the objective function. We compare the proposed method with several 2D unsupervised dimensionality methods. K-means is used to evaluate the clustering performance. We conduct extensive experiments on Coil20, AT&T, FERET, USPS, and Yale data sets to verify the effectiveness of our proposed method.

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50. Multiassociative Memory: Recurrent Synapses Increase Storage Capacity

Marcelo Matheus Gauy, Florian Meier, Angelika Steger

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The connection density of nearby neurons in the cortex has been observed to be around 0.1, whereas the longer-range connections are present with much sparser density (Kalisman, Silberberg, & Markram, 2005). We propose a memory association model that qualitatively explains these empirical observations. The model we consider is a multiassociative, sparse, Willshaw-like model consisting of binary threshold neurons and binary synapses. It uses recurrent synapses for iterative retrieval of stored memories. We quantify the usefulness of recurrent synapses by simulating the model for small network sizes and by doing a precise mathematical analysis for large network sizes. Given the network parameters, we can determine the precise values of recurrent and afferent synapse densities that optimize the storage capacity of the network. If the network size is like that of a cortical column, then the predicted optimal recurrent density lies in a range that is compatible with biological measurements. Furthermore, we show that our model is able to surpass the standard Willshaw model in the multiassociative case if the information capacity is normalized per strong synapse or per bits required to store the model, as considered in Knoblauch, Palm, and Sommer (2010).

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51. DC Algorithm for Extended Robust Support Vector Machine

Shuhei Fujiwara, Akiko Takeda, Takafumi Kanamori

https://doi.org/10.1162/neco_a_00958 | pp. 1406-1438

Nonconvex variants of support vector machines (SVMs) have been developed for various purposes. For example, robust SVMs attain robustness to outliers by using a nonconvex loss function, while extended [Formula: see text]-SVM (E[Formula: see text]-SVM) extends the range of the hyperparameter by introducing a nonconvex constraint. Here, we consider an extended robust support vector machine (ER-SVM), a robust variant of E[Formula: see text]-SVM. ER-SVM combines two types of nonconvexity from robust SVMs and E[Formula: see text]-SVM. Because of the two nonconvexities, the existing algorithm we proposed needs to be divided into two parts depending on whether the hyperparameter value is in the extended range or not. The algorithm also heuristically solves the nonconvex problem in the extended range. In this letter, we propose a new, efficient algorithm for ER-SVM. The algorithm deals with two types of nonconvexity while never entailing more computations than either E[Formula: see text]-SVM or robust SVM, and it finds a critical point of ER-SVM. Furthermore, we show that ER-SVM includes the existing robust SVMs as special cases. Numerical experiments confirm the effectiveness of integrating the two nonconvexities.

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52. Erratum to “A Note on Divergences”

Xiao Liang

https://doi.org/10.1162/neco_e_00960 | pp. 1439-1439

Erratum to “A Note on Divergences” by Xiao Liang, Neural Computation Vol. 28, No. 10, pp. 2045–2062 (2016)doi:10.1162/NECO_a_00878In order to assist readers interested in obtaining new results on divergences, I write this erratum to the version of this article initially published.The second paragraph of Remark 1 should appear as follows:We just disproved Amari’s conjecture for the general case of in . It is still

interesting and significant to study whether Amari's conjecture is true or not for and some nontrivial subset of , such as the set of all probability measures on .“” in equation 4.5 should be “.”On page 2058, “” in line 2 should be “.”Acknowledgments should appear as follows:I thank the referees very much for helpful comments and suggestions, which led to my revision on section 3. Moreover, I am grateful to the referees for bringing Amari (2016) and Jiao, Courtade, No, Venkat, and Weissman (2015) to my attention. I thank Z. Lei of Fudan University for valuable instructions, comments, and suggestions on the note.On page 2062, “Interference” in line 2 should be “Statistics.”

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53. A Unified Theory of Neuro-MRI Data Shows Scale-Free Nature of Connectivity Modes

V. L. Galinsky, Lawrence R. Frank

https://doi.org/10.1162/neco_a_00955 | pp. 1441-1467

A primary goal of many neuroimaging studies that use magnetic resonance imaging (MRI) is to deduce the structure-function relationships in the human brain using data from the three major neuro-MRI modalities: high-resolution anatomical, diffusion tensor imaging, and functional MRI. To date, the general procedure for analyzing these data is to combine the results derived independently from each of these modalities. In this article, we develop a new theoretical and computational approach for combining these different MRI modalities into a powerful and versatile framework that combines our recently developed methods for morphological shape analysis and segmentation, simultaneous local diffusion estimation and global tractography, and nonlinear and nongaussian spatial-temporal activation pattern classification and ranking, as well as our fast and accurate approach for nonlinear registration between modalities. This joint analysis method is capable of extracting new levels of information that is not achievable from any of those single modalities alone. A theoretical probabilistic framework based on a reformulation of prior information and available interdependencies between modalities through a joint coupling matrix and an efficient computational implementation allows construction of quantitative functional, structural, and effective brain connectivity modes and parcellation. This new method provides an overall increase of resolution, accuracy, level of detail, and information content and has the potential to be instrumental in the clinical adaptation of neuro-MRI modalities, which, when jointly analyzed, provide a more comprehensive view of a subject's structure-function relations, while the current standard, wherein single-modality methods are analyzed separately, leaves a critical gap in an integrated view of a subject's neuorphysiological state. As one example of this increased sensitivity, we demonstrate that the jointly estimated structural and functional dependencies of mode power follow the same power law decay with the same exponent.

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54. Mean First Passage Memory Lifetimes by Reducing Complex Synapses to Simple Synapses

Terry Elliott

https://doi.org/10.1162/neco_a_00956 | pp. 1468-1527

Memory models that store new memories by forgetting old ones have memory lifetimes that are rather short and grow only logarithmically in the number of synapses. Attempts to overcome these deficits include "complex" models of synaptic plasticity in which synapses possess internal states governing the expression of synaptic plasticity. Integrate-and-express, filter-based models of synaptic plasticity propose that synapses

act as low-pass filters, integrating plasticity induction signals before expressing synaptic plasticity. Such mechanisms enhance memory lifetimes, leading to an initial rise in the memory signal that is in radical contrast to other related, but nonintegrative, memory models. Because of the complexity of models with internal synaptic states, however, their dynamics can be more difficult to extract compared to "simple" models that lack internal states. Here, we show that by focusing only on processes that lead to changes in synaptic strength, we can integrate out internal synaptic states and effectively reduce complex synapses to simple synapses. For binary-strength synapses, these simplified dynamics then allow us to work directly in the transitions in perceptron activation induced by memory storage rather than in the underlying transitions in synaptic configurations. This permits us to write down master and Fokker-Planck equations that may be simplified under certain, well-defined approximations. These methods allow us to see that memory based on synaptic filters can be viewed as an initial transient that leads to memory signal rise, followed by the emergence of Ornstein-Uhlenbeck-like dynamics that return the system to equilibrium. We may use this approach to compute mean first passage time-defined memory lifetimes for complex models of memory storage.

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55. Information Rate Analysis of a Synaptic Release Site Using a Two-State Model of Short-Term Depression

Mehrdad Salmasi, Martin Stemmler, Stefan Glasauer, Alex Loebel

https://doi.org/10.1162/neco_a_00962 | pp. 1528-1560

Synapses are the communication channels for information transfer between neurons; these are the points at which pulse-like signals are converted into the stochastic release of quantized amounts of chemical neurotransmitter. At many synapses, prior neuronal activity depletes synaptic resources, depressing subsequent responses of both spontaneous and spike-evoked releases. We analytically compute the information transmission rate of a synaptic release site, which we model as a binary asymmetric channel. Short-term depression is incorporated by assigning the channel a memory of depth one. A successful release, whether spike evoked or spontaneous, decreases the probability of a subsequent release; if no release occurs on the following time step, the release probabilities recover back to their default values. We prove that synaptic depression can increase the release site's information rate if spontaneous release is more strongly depressed than spike-evoked release. When depression affects spontaneous and evoked release equally, the information rate must invariably decrease, even when the rate is normalized by the resources used for synaptic transmission. For identical depression levels, we analytically disprove the hypothesis, at least in this simplified model, that synaptic depression serves energy- and information-efficient encoding.

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56. Evidence Accumulation and Change Rate Inference in Dynamic Environments

Adrian E. Radillo, Alan Veliz■Cuba, Krešimir Josi■■, Zachary P. Kilpatrick

https://doi.org/10.1162/neco_a_00957 | pp. 1561-1610

In a constantly changing world, animals must account for environmental volatility when making decisions. To appropriately discount older, irrelevant information, they need to learn the rate at which the environment changes. We develop an ideal observer model capable of inferring the present state of the environment

along with its rate of change. Key to this computation is an update of the posterior probability of all possible change point counts. This computation can be challenging, as the number of possibilities grows rapidly with time. However, we show how the computations can be simplified in the continuum limit by a moment closure approximation. The resulting low-dimensional system can be used to infer the environmental state and change rate with accuracy comparable to the ideal observer. The approximate computations can be performed by a neural network model via a rate-correlation-based plasticity rule. We thus show how optimal observers accumulate evidence in changing environments and map this computation to reduced models that perform inference using plausible neural mechanisms.

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57. The Deterministic Information Bottleneck

DJ Strouse, David J. Schwab

https://doi.org/10.1162/neco_a_00961 | pp. 1611-1630

Lossy compression and clustering fundamentally involve a decision about which features are relevant and which are not. The information bottleneck method (IB) by Tishby, Pereira, and Bialek (1999) formalized this notion as an information-theoretic optimization problem and proposed an optimal trade-off between throwing away as many bits as possible and selectively keeping those that are most important. In the IB, compression is measured by mutual information. Here, we introduce an alternative formulation that replaces mutual information with entropy, which we call the deterministic information bottleneck (DIB) and argue better captures this notion of compression. As suggested by its name, the solution to the DIB problem turns out to be a deterministic encoder, or hard clustering, as opposed to the stochastic encoder, or soft clustering, that is optimal under the IB. We compare the IB and DIB on synthetic data, showing that the IB and DIB perform similarly in terms of the IB cost function, but that the DIB significantly outperforms the IB in terms of the DIB cost function. We also empirically find that the DIB offers a considerable gain in computational efficiency over the IB, over a range of convergence parameters. Our derivation of the DIB also suggests a method for continuously interpolating between the soft clustering of the IB and the hard clustering of the DIB.

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58. Robust Averaging of Covariances for EEG Recordings Classification in Motor Imagery Brain-Computer Interfaces

Takashi Uehara, Matteo Sartori, Toshihisa Tanaka, Simone Fiori

https://doi.org/10.1162/neco_a_00963 | pp. 1631-1666

The estimation of covariance matrices is of prime importance to analyze the distribution of multivariate signals. In motor imagery-based brain-computer interfaces (MI-BCI), covariance matrices play a central role in the extraction of features from recorded electroencephalograms (EEGs); therefore, correctly estimating covariance is crucial for EEG classification. This letter discusses algorithms to average sample covariance matrices (SCMs) for the selection of the reference matrix in tangent space mapping (TSM)-based MI-BCI. Tangent space mapping is a powerful method of feature extraction and strongly depends on the selection of a reference covariance matrix. In general, the observed signals may include outliers; therefore, taking the geometric mean of SCMs as the reference matrix may not be the best choice. In order to deal with the effects of outliers, robust estimators have to be used. In particular, we discuss and test the use of geometric medians and trimmed averages (defined on the basis of several metrics) as robust estimators. The main idea behind trimmed averages is to eliminate data that exhibit the largest distance from the average

covariance calculated on the basis of all available data. The results of the experiments show that while the geometric medians show little differences from conventional methods in terms of classification accuracy in the classification of electroencephalographic recordings, the trimmed averages show significant improvement for all subjects.

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59. Olfactory Recognition Based on EEG Gamma-Band Activity

Önder Aydemir

https://doi.org/10.1162/neco_a_00966 | pp. 1667-1680

There are various kinds of brain monitoring techniques, including local field potential, near-infrared spectroscopy, magnetic resonance imaging (MRI), positron emission tomography, functional MRI, electroencephalography (EEG), and magnetoencephalography. Among those techniques, EEG is the most widely used one due to its portability, low setup cost, and noninvasiveness. Apart from other advantages, EEG signals also help to evaluate the ability of the smelling organ. In such studies, EEG signals, which are recorded during smelling, are analyzed to determine the subject lacks any smelling ability or to measure the response of the brain. The main idea of this study is to show the emotional difference in EEG signals during perception of valerian, lotus flower, cheese, and rosewater odors by the EEG gamma wave. The proposed method was applied to the EEG signals, which were taken from five healthy subjects in the conditions of eyes open and eyes closed at the Swiss Federal Institute of Technology. In order to represent the signals, we extracted features from the gamma band of the EEG trials by continuous wavelet transform with the selection of Morlet as a wavelet function. Then the [Formula: see text]-nearest neighbor algorithm was implemented as the classifier for recognizing the EEG trials as valerian, lotus flower, cheese, and rosewater. We achieved an average classification accuracy rate of 87.50% with the 4.3 standard deviation value for the subjects in eyes-open condition and an average classification accuracy rate of 94.12% with the 2.9 standard deviation value for the subjects in eyes-closed condition. The results prove that the proposed continuous wavelet transform-based feature extraction method has great potential to classify the EEG signals recorded during smelling of the present odors. It has been also established that gamma-band activity of the brain is highly associated with olfaction.

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60. Nested-Clique Network Model of Neural Associative Memory

Asieh Abolpour Mofrad, Matthew G. Parker

https://doi.org/10.1162/neco_a_00964 | pp. 1681-1695

Clique-based neural associative memories introduced by Gripon and Berrou (GB), have been shown to have good performance, and in our previous work we improved the learning capacity and retrieval rate by local coding and precoding in the presence of partial erasures. We now take a step forward and consider nested-clique graph structures for the network. The GB model stores patterns as small cliques, and we here replace these by nested cliques. Simulation results show that the nested-clique structure enhances the clique-based model.

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61. Chaotic Pattern Alternations Can Reproduce Properties of Dominance Durations in Multistable Perception

Takashi Kanamaru

https://doi.org/10.1162/neco_a_00965 | pp. 1696-1720

We propose a pulse neural network that exhibits chaotic pattern alternations among stored patterns as a model of multistable perception, which is reflected in phenomena such as binocular rivalry and perceptual ambiguity. When we regard the mixed state of patterns as a part of each pattern, the durations of the retrieved pattern obey unimodal distributions. We confirmed that no chaotic properties are observed in the time series of durations, consistent with the findings of previous psychological studies. Moreover, it is shown that our model also reproduces two properties of multistable perception that characterize the relationship between the contrast of inputs and the durations.

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62. Lag Synchronization Criteria for Memristor-Based Coupled Neural Networks via Parameter Mismatches Analysis Approach

Ning Li, Jinde Cao, Ahmed Alsaedi, Fuad E. Alsaadi

https://doi.org/10.1162/neco_a_00918 | pp. 1721-1744

This letter focuses on lag synchronization control analysis for memristor-based coupled neural networks with parameter mismatches. Due to the parameter mismatches, lag complete synchronization in general cannot be achieved. First, based on the [Formula: see text]-measure method, generalized Halanay inequality, together with control algorithms, some sufficient conditions are obtained to ensure that coupled memristor-based neural networks are in a state of lag synchronization with an error. Moreover, the error level is estimated. Second, we show that memristor-based coupled neural networks with parameter mismatches can reach lag complete synchronization under a discontinuous controller. Finally, two examples are given to illustrate the effectiveness of the proposed criteria and well support theoretical results.

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63. Extraction of Synaptic Input Propertiesin Vivo

Paolo Puggioni, Márta Jelitai, Ian Duguid, Mark C. W. van Rossum

https://doi.org/10.1162/neco_a_00975 | pp. 1745-1768

Knowledge of synaptic input is crucial for understanding synaptic integration and ultimately neural function. However, in vivo, the rates at which synaptic inputs arrive are high, so that it is typically impossible to detect single events. We show here that it is nevertheless possible to extract the properties of the events and, in particular, to extract the event rate, the synaptic time constants, and the properties of the event size distribution from in vivo voltage-clamp recordings. Applied to cerebellar interneurons, our method reveals that the synaptic input rate increases from 600 Hz during rest to 1000 Hz during locomotion, while the amplitude and shape of the synaptic events are unaffected by this state change. This method thus complements existing methods to measure neural function in vivo.

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64. Effect of Phase Response Curve Shape and Synaptic Driving Force on Synchronization of Coupled Neuronal Oscillators

Ramana Dodla, Charles J. Wilson
https://doi.org/10.1162/neco_a_00978 | pp. 1769-1814

The role of the phase response curve (PRC) shape on the synchrony of synaptically coupled oscillating neurons is examined. If the PRC is independent of the phase, because of the synaptic form of the coupling, synchrony is found to be stable for both excitatory and inhibitory coupling at all rates, whereas the antisynchrony becomes stable at low rates. A faster synaptic rise helps extend the stability of antisynchrony to higher rates. If the PRC is not constant but has a profile like that of a leaky integrate-and-fire model, then, in contrast to the earlier reports that did not include the voltage effects, mutual excitation could lead to stable synchrony provided the synaptic reversal potential is below the voltage level the neuron would have reached in the absence of the interaction and threshold reset. This level is controlled by the applied current and the leakage parameters. Such synchrony is contingent on significant phase response (that would result, for example, by a sharp PRC jump) occurring during the synaptic rising phase. The rising phase, however, does not contribute significantly if it occurs before the voltage spike reaches its peak. Then a stable near-synchronous state can still exist between type 1 PRC neurons if the PRC shows a left skewness in its shape. These results are examined comprehensively using perfect integrate-and-fire, leaky integrate-and-fire, and skewed PRC shapes under the assumption of the weakly coupled oscillator theory applied to synaptically coupled neuron models.

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65. Determination of the Lowest-Energy States for the Model Distribution of Trained Restricted Boltzmann Machines Using a 1000 Qubit D-Wave 2X Quantum Computer

Yaroslav Koshka, Dilina Perera, Spencer R. Hall, M. A. Novotny
https://doi.org/10.1162/neco_a_00974 | pp. 1815-1837

The possibility of using a quantum computer D-Wave 2X with more than 1000 qubits to determine the global minimum of the energy landscape of trained restricted Boltzmann machines is investigated. In order to overcome the problem of limited interconnectivity in the D-Wave architecture, the proposed RBM embedding combines multiple qubits to represent a particular RBM unit. The results for the lowest-energy (the ground state) and some of the higher-energy states found by the D-Wave 2X were compared with those of the classical simulated annealing (SA) algorithm. In many cases, the D-Wave machine successfully found the same RBM lowest-energy state as that found by SA. In some examples, the D-Wave machine returned a state corresponding to one of the higher-energy local minima found by SA. The inherently nonperfect embedding of the RBM into the Chimera lattice explored in this work (i.e., multiple qubits combined into a single RBM unit were found not to be guaranteed to be all aligned) and the existence of small, persistent biases in the D-Wave hardware may cause a discrepancy between the D-Wave and the SA results. In some of the investigated cases, introduction of a small bias field into the energy function or optimization of the chain-strength parameter in the D-Wave embedding successfully addressed difficulties of the particular RBM embedding. With further development of the D-Wave hardware, the approach will be suitable for much larger numbers of RBM units.

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66. Local Intrinsic Dimension Estimation by Generalized Linear Modeling

Hideitsu Hino, Jun Fujiki, Shotaro Akaho, Noboru Murata

https://doi.org/10.1162/neco_a_00969 | pp. 1838-1878

We propose a method for intrinsic dimension estimation. By fitting the power of distance from an inspection point and the number of samples included inside a ball with a radius equal to the distance, to a regression model, we estimate the goodness of fit. Then, by using the maximum likelihood method, we estimate the local intrinsic dimension around the inspection point. The proposed method is shown to be comparable to conventional methods in global intrinsic dimension estimation experiments. Furthermore, we experimentally show that the proposed method outperforms a conventional local dimension estimation method.

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67. Generalization Analysis of Fredholm Kernel Regularized Classifiers

Tieliang Gong, Zongben Xu, Hong Chen

https://doi.org/10.1162/neco_a_00967 | pp. 1879-1901

Recently, a new framework, Fredholm learning, was proposed for semisupervised learning problems based on solving a regularized Fredholm integral equation. It allows a natural way to incorporate unlabeled data into learning algorithms to improve their prediction performance. Despite rapid progress on implementable algorithms with theoretical guarantees, the generalization ability of Fredholm kernel learning has not been studied. In this letter, we focus on investigating the generalization performance of a family of classification algorithms, referred to as Fredholm kernel regularized classifiers. We prove that the corresponding learning rate can achieve [Formula: see text] ([Formula: see text] is the number of labeled samples) in a limiting case. In addition, a representer theorem is provided for the proposed regularized scheme, which underlies its applications.

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68. A Weight-Adaptive Laplacian Embedding for Graph-Based Clustering

De Cheng, Feiping Nie, Jiande Sun, Yihong Gong

https://doi.org/10.1162/neco_a_00973 | pp. 1902-1918

Graph-based clustering methods perform clustering on a fixed input data graph. Thus such clustering results are sensitive to the particular graph construction. If this initial construction is of low quality, the resulting clustering may also be of low quality. We address this drawback by allowing the data graph itself to be adaptively adjusted in the clustering procedure. In particular, our proposed weight adaptive Laplacian (WAL) method learns a new data similarity matrix that can adaptively adjust the initial graph according to the similarity weight in the input data graph. We develop three versions of these methods based on the L2-norm, fuzzy entropy regularizer, and another exponential-based weight strategy, that yield three new graph-based clustering objectives. We derive optimization algorithms to solve these objectives. Experimental results on synthetic data sets and real-world benchmark data sets exhibit the effectiveness of these new graph-based clustering methods.

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69. Support Vector Algorithms for Optimizing the Partial Area under the ROC Curve

Harikrishna Narasimhan, Shivani Agarwal

https://doi.org/10.1162/neco_a_00972 | pp. 1919-1963

The area under the ROC curve (AUC) is a widely used performance measure in machine learning. Increasingly, however, in several applications, ranging from ranking to biometric screening to medicine, performance is measured not in terms of the full area under the ROC curve but in terms of the partial area under the ROC curve between two false-positive rates. In this letter, we develop support vector algorithms for directly optimizing the partial AUC between any two false-positive rates. Our methods are based on minimizing a suitable proxy or surrogate objective for the partial AUC error. In the case of the full AUC, one can readily construct and optimize convex surrogates by expressing the performance measure as a summation of pairwise terms. The partial AUC, on the other hand, does not admit such a simple decomposable structure, making it more challenging to design and optimize (tight) convex surrogates for this measure. Our approach builds on the structural SVM framework of Joachims (2005) to design convex surrogates for partial AUC and solves the resulting optimization problem using a cutting plane solver. Unlike the full AUC, where the combinatorial optimization needed in each iteration of the cutting plane solver can be decomposed and solved efficiently, the corresponding problem for the partial AUC is harder to decompose. One of our main contributions is a polynomial time algorithm for solving the combinatorial optimization problem associated with partial AUC. We also develop an approach for optimizing a tighter nonconvex hinge loss-based surrogate for the partial AUC using difference-of-convex programming. Our experiments on a variety of real-world and benchmark tasks confirm the efficacy of the proposed methods.

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70. A Customized Attention-Based Long Short-Term Memory Network for Distant Supervised Relation Extraction

Dengchao He, Hongjun Zhang, Wenning Hao, Rui Zhang, Kai Cheng

https://doi.org/10.1162/neco_a_00970 | pp. 1964-1985

Distant supervision, a widely applied approach in the field of relation extraction can automatically generate large amounts of labeled training corpus with minimal manual effort. However, the labeled training corpus may have many false-positive data, which would hurt the performance of relation extraction. Moreover, in traditional feature-based distant supervised approaches, extraction models adopt human design features with natural language processing. It may also cause poor performance. To address these two shortcomings, we propose a customized attention-based long short-term memory network. Our approach adopts word-level attention to achieve better data representation for relation extraction without manually designed features to perform distant supervision instead of fully supervised relation extraction, and it utilizes instance-level attention to tackle the problem of false-positive data. Experimental results demonstrate that our proposed approach is effective and achieves better performance than traditional methods.

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71. Multiview Feature Analysis via Structured Sparsity and Shared Subspace Discovery

Yan■Shuo Chang, Feiping Nie, Wang Ming-yu

https://doi.org/10.1162/neco_a_00977 | pp. 1986-2003

Since combining features from heterogeneous data sources can significantly boost classification performance in many applications, it has attracted much research attention over the past few years. Most of the existing multiview feature analysis approaches separately learn features in each view, ignoring knowledge shared by multiple views. Different views of features may have some intrinsic correlations that

might be beneficial to feature learning. Therefore, it is assumed that multiviews share subspaces from which common knowledge can be discovered. In this letter, we propose a new multiview feature learning algorithm, aiming to exploit common features shared by different views. To achieve this goal, we propose a feature learning algorithm in a batch mode, by which the correlations among different views are taken into account. Multiple transformation matrices for different views are simultaneously learned in a joint framework. In this way, our algorithm can exploit potential correlations among views as supplementary information that further improves the performance result. Since the proposed objective function is nonsmooth and difficult to solve directly, we propose an iterative algorithm for effective optimization. Extensive experiments have been conducted on a number of real-world data sets. Experimental results demonstrate superior performance in terms of classification against all the compared approaches. Also, the convergence guarantee has been validated in the experiment.

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72. Analytical Derivation of Nonlinear Spectral Effects and 1/f Scaling Artifact in Signal Processing of Real-World Data

Claudia Lainscsek, Lyle Muller, Aaron L. Sampson, Terrence J. Sejnowski

https://doi.org/10.1162/neco_a_00979 | pp. 2004-2020

In estimating the frequency spectrum of real-world time series data, we must violate the assumption of infinite-length, orthogonal components in the Fourier basis. While it is widely known that care must be taken with discretely sampled data to avoid aliasing of high frequencies, less attention is given to the influence of low frequencies with period below the sampling time window. Here, we derive an analytic expression for the side-lobe attenuation of signal components in the frequency domain representation. This expression allows us to detail the influence of individual frequency components throughout the spectrum. The first consequence is that the presence of low-frequency components introduces a 1/f[Formula: see text] component across the power spectrum, with a scaling exponent of [Formula: see text]. This scaling artifact could be composed of diffuse low-frequency components, which can render it difficult to detect a priori. Further, treatment of the signal with standard digital signal processing techniques cannot easily remove this scaling component. While several theoretical models have been introduced to explain the ubiquitous 1/f[Formula: see text] scaling component in neuroscientific data, we conjecture here that some experimental observations could be the result of such data analysis procedures.

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73. Population Vectors Can Provide Near Optimal Integration of Information

Josue Orellana, Jordan Rodu, Robert E. Kass

https://doi.org/10.1162/neco_a_00992 | pp. 2021-2029

Much attention has been paid to the question of how Bayesian integration of information could be implemented by a simple neural mechanism. We show that population vectors based on point-process inputs combine evidence in a form that closely resembles Bayesian inference, with each input spike carrying information about the tuning of the input neuron. We also show that population vectors can combine information relatively accurately in the presence of noisy synaptic encoding of tuning curves.

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74. Predicting the Existence and Stability of Phase-Locked Mode in Neural Networks Using Generalized Phase-Resetting Curve

Sorinel A. Oprisan

https://doi.org/10.1162/neco_a_00983 | pp. 2030-2054

We used the phase-resetting method to study a biologically relevant three-neuron network in which one neuron receives multiple inputs per cycle. For this purpose, we first generalized the concept of phase resetting to accommodate multiple inputs per cycle. We explicitly showed how analytical conditions for the existence and the stability of phase-locked modes are derived. In particular, we solved newly derived recursive maps using as an example a biologically relevant driving-driven neural network with a dynamic feedback loop. We applied the generalized phase-resetting definition to predict the relative-phase and the stability of a phase-locked mode in open loop setup. We also compared the predicted phase-locked mode against numerical simulations of the fully connected network.

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75. Multivariate Time Series Decomposition into Oscillation Components

Takeru Matsuda, Fumiyasu Komaki

https://doi.org/10.1162/neco_a_00981 | pp. 2055-2075

Many time series are considered to be a superposition of several oscillation components. We have proposed a method for decomposing univariate time series into oscillation components and estimating their phases (Matsuda & Komaki, 2017). In this study, we extend that method to multivariate time series. We assume that several oscillators underlie the given multivariate time series and that each variable corresponds to a superposition of the projections of the oscillators. Thus, the oscillators superpose on each variable with amplitude and phase modulation. Based on this idea, we develop gaussian linear state-space models and use them to decompose the given multivariate time series. The model parameters are estimated from data using the empirical Bayes method, and the number of oscillators is determined using the Akaike information criterion. Therefore, the proposed method extracts underlying oscillators in a data-driven manner and enables investigation of phase dynamics in a given multivariate time series. Numerical results show the effectiveness of the proposed method. From monthly mean north-south sunspot number data, the proposed method reveals an interesting phase relationship.

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76. Direct Estimation of the Derivative of Quadratic Mutual Information with Application in Supervised Dimension Reduction

Voot Tangkaratt, Hiroaki Sasaki, Masashi Sugiyama

https://doi.org/10.1162/neco_a_00986 | pp. 2076-2122

A typical goal of linear-supervised dimension reduction is to find a low-dimensional subspace of the input space such that the projected input variables preserve maximal information about the output variables. The dependence-maximization approach solves the supervised dimension-reduction problem through maximizing a statistical dependence between projected input variables and output variables. A well-known statistical dependence measure is mutual information (MI), which is based on the Kullback-Leibler (KL) divergence. However, it is known that the KL divergence is sensitive to outliers. Quadratic MI (QMI) is a variant of MI based on the [Formula: see text] distance, which is more robust against outliers than the KL

divergence, and a computationally efficient method to estimate QMI from data, least squares QMI (LSQMI), has been proposed recently. For these reasons, developing a supervised dimension-reduction method based on LSQMI seems promising. However, not QMI itself but the derivative of QMI is needed for subspace search in linear-supervised dimension reduction, and the derivative of an accurate QMI estimator is not necessarily a good estimator of the derivative of QMI. In this letter, we propose to directly estimate the derivative of QMI without estimating QMI itself. We show that the direct estimation of the derivative of QMI is more accurate than the derivative of the estimated QMI. Finally, we develop a linear-supervised dimension-reduction algorithm that efficiently uses the proposed derivative estimator and demonstrate through experiments that the proposed method is more robust against outliers than existing methods.

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77. Deep Restricted Kernel Machines Using Conjugate Feature Duality

Johan A. K. Suykens

https://doi.org/10.1162/neco_a_00984 | pp. 2123-2163

The aim of this letter is to propose a theory of deep restricted kernel machines offering new foundations for deep learning with kernel machines. From the viewpoint of deep learning, it is partially related to restricted Boltzmann machines, which are characterized by visible and hidden units in a bipartite graph without hidden-to-hidden connections and deep learning extensions as deep belief networks and deep Boltzmann machines. From the viewpoint of kernel machines, it includes least squares support vector machines for classification and regression, kernel principal component analysis (PCA), matrix singular value decomposition, and Parzen-type models. A key element is to first characterize these kernel machines in terms of so-called conjugate feature duality, yielding a representation with visible and hidden units. It is shown how this is related to the energy form in restricted Boltzmann machines, with continuous variables in a nonprobabilistic setting. In this new framework of so-called restricted kernel machine (RKM) representations, the dual variables correspond to hidden features. Deep RKM are obtained by coupling the RKMs. The method is illustrated for deep RKM, consisting of three levels with a least squares support vector machine regression level and two kernel PCA levels. In its primal form also deep feedforward neural networks can be trained within this framework.

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78. Rank Selection in Nonnegative Matrix Factorization using Minimum Description Length

Steven Squires, Adam Prügel■Bennett, Mahesan Niranjan

https://doi.org/10.1162/neco_a_00980 | pp. 2164-2176

Nonnegative matrix factorization (NMF) is primarily a linear dimensionality reduction technique that factorizes a nonnegative data matrix into two smaller nonnegative matrices: one that represents the basis of the new subspace and the second that holds the coefficients of all the data points in that new space. In principle, the nonnegativity constraint forces the representation to be sparse and parts based. Instead of extracting holistic features from the data, real parts are extracted that should be significantly easier to interpret and analyze. The size of the new subspace selects how many features will be extracted from the data. An effective choice should minimize the noise while extracting the key features. We propose a mechanism for selecting the subspace size by using a minimum description length technique. We demonstrate that our technique provides plausible estimates for real data as well as accurately predicting

the known size of synthetic data. We provide an implementation of our code in a Matlab format.

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79. GP-Select: Accelerating EM Using Adaptive Subspace Preselection

Jacquelyn A. Shelton, Jan Gasthaus, Zhenwen Dai, Jörg Lücke, Arthur Gretton
https://doi.org/10.1162/neco_a_00982 | pp. 2177-2202

We propose a nonparametric procedure to achieve fast inference in generative graphical models when the number of latent states is very large. The approach is based on iterative latent variable preselection, where we alternate between learning a selection function to reveal the relevant latent variables and using this to obtain a compact approximation of the posterior distribution for EM. This can make inference possible where the number of possible latent states is, for example, exponential in the number of latent variables, whereas an exact approach would be computationally infeasible. We learn the selection function entirely from the observed data and current expectation-maximization state via gaussian process regression. This is in contrast to earlier approaches, where selection functions were manually designed for each problem setting. We show that our approach performs as well as these bespoke selection functions on a wide variety of inference problems. In particular, for the challenging case of a hierarchical model for object localization with occlusion, we achieve results that match a customized state-of-the-art selection method at a far lower computational cost.

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80. LQG Online Learning

Giorgio Gnecco, Alberto Bemporad, Marco Gori, Marcello Sanguineti
https://doi.org/10.1162/neco_a_00976 | pp. 2203-2291

Optimal control theory and machine learning techniques are combined to formulate and solve in closed form an optimal control formulation of online learning from supervised examples with regularization of the updates. The connections with the classical linear quadratic gaussian (LQG) optimal control problem, of which the proposed learning paradigm is a nontrivial variation as it involves random matrices, are investigated. The obtained optimal solutions are compared with the Kalman filter estimate of the parameter vector to be learned. It is shown that the proposed algorithm is less sensitive to outliers with respect to the Kalman estimate (thanks to the presence of the regularization term), thus providing smoother estimates with respect to time. The basic formulation of the proposed online learning framework refers to a discrete-time setting with a finite learning horizon and a linear model. Various extensions are investigated, including the infinite learning horizon and, via the so-called kernel trick, the case of nonlinear models.

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81. Sensory Metrics of Neuromechanical Trust

William Softky, Criscillia Benford
https://doi.org/10.1162/neco_a_00988 | pp. 2293-2351

Today digital sources supply a historically unprecedented component of human sensorimotor data, the consumption of which is correlated with poorly understood maladies such as Internet addiction disorder and Internet gaming disorder. Because both natural and digital sensorimotor data share common mathematical descriptions, one can quantify our informational sensorimotor needs using the signal processing metrics of

entropy, noise, dimensionality, continuity, latency, and bandwidth. Such metrics describe in neutral terms the informational diet human brains require to self-calibrate, allowing individuals to maintain trusting relationships. With these metrics, we define the trust humans experience using the mathematical language of computational models, that is, as a primitive statistical algorithm processing finely grained sensorimotor data from neuromechanical interaction. This definition of neuromechanical trust implies that artificial sensorimotor inputs and interactions that attract low-level attention through frequent discontinuities and enhanced coherence will decalibrate a brain's representation of its world over the long term by violating the implicit statistical contract for which self-calibration evolved. Our hypersimplified mathematical understanding of human sensorimotor processing as multiscale, continuous-time vibratory interaction allows equally broad-brush descriptions of failure modes and solutions. For example, we model addiction in general as the result of homeostatic regulation gone awry in novel environments (sign reversal) and digital dependency as a sub-case in which the decalibration caused by digital sensorimotor data spurs yet more consumption of them. We predict that institutions can use these sensorimotor metrics to quantify media richness to improve employee well-being; that dyads and family-size groups will bond and heal best through low-latency, high-resolution multisensory interaction such as shared meals and reciprocated touch; and that individuals can improve sensory and sociosensory resolution through deliberate sensory reintegration practices. We conclude that we humans are the victims of our own success, our hands so skilled they fill the world with captivating things, our eyes so innocent they follow eagerly.

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82. Deep Convolutional Neural Networks for Image Classification: A Comprehensive Review

Waseem Rawat, Zenghui Wang

https://doi.org/10.1162/neco_a_00990 | pp. 2352-2449

Convolutional neural networks (CNNs) have been applied to visual tasks since the late 1980s. However, despite a few scattered applications, they were dormant until the mid-2000s when developments in computing power and the advent of large amounts of labeled data, supplemented by improved algorithms, contributed to their advancement and brought them to the forefront of a neural network renaissance that has seen rapid progression since 2012. In this review, which focuses on the application of CNNs to image classification tasks, we cover their development, from their predecessors up to recent state-of-the-art deep learning systems. Along the way, we analyze (1) their early successes, (2) their role in the deep learning renaissance, (3) selected symbolic works that have contributed to their recent popularity, and (4) several improvement attempts by reviewing contributions and challenges of over 300 publications. We also introduce some of their current trends and remaining challenges.

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83. Implementing a Bayes Filter in a Neural Circuit: The Case of Unknown Stimulus Dynamics

Sacha Sokoloski

https://doi.org/10.1162/neco_a_00991 | pp. 2450-2490

In order to interact intelligently with objects in the world, animals must first transform neural population responses into estimates of the dynamic, unknown stimuli that caused them. The Bayesian solution to this problem is known as a Bayes filter, which applies Bayes' rule to combine population responses with the

predictions of an internal model. The internal model of the Bayes filter is based on the true stimulus dynamics, and in this note, we present a method for training a theoretical neural circuit to approximately implement a Bayes filter when the stimulus dynamics are unknown. To do this we use the inferential properties of linear probabilistic population codes to compute Bayes' rule and train a neural network to compute approximate predictions by the method of maximum likelihood. In particular, we perform stochastic gradient descent on the negative log-likelihood of the neural network parameters with a novel approximation of the gradient. We demonstrate our methods on a finite-state, a linear, and a nonlinear filtering problem and show how the hidden layer of the neural network develops tuning curves consistent with findings in experimental neuroscience.

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84. Cortical Spike Synchrony as a Measure of Input Familiarity

Clemens Korndörfer, Ekkehard Ullner, Jordi García-Ojalvo, Gordon Pipa

https://doi.org/10.1162/neco_a_00987 | pp. 2491-2510

Spike synchrony, which occurs in various cortical areas in response to specific perception, action, and memory tasks, has sparked a long-standing debate on the nature of temporal organization in cortex. One prominent view is that this type of synchrony facilitates the binding or grouping of separate stimulus components. We argue instead for a more general function: a measure of the prior probability of incoming stimuli, implemented by long-range, horizontal, intracortical connections. We show that networks of this kind-pulse-coupled excitatory spiking networks in a noisy environment-can provide a sufficient substrate for stimulus-dependent spike synchrony. This allows for a quick (few spikes) estimate of the match between inputs and the input history as encoded in the network structure. Given the ubiquity of small, strongly excitatory subnetworks in cortex, we thus propose that many experimental observations of spike synchrony can be viewed as signs of input patterns that resemble long-term experience-that is, of patterns with high prior probability.

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85. Dendrites Enable a Robust Mechanism for Neuronal Stimulus Selectivity

Romain D. Cazé, Sarah Jarvis, Amanda J. Foust, Simon R. Schultz

https://doi.org/10.1162/neco_a_00989 | pp. 2511-2527

Hearing, vision, touch: underlying all of these senses is stimulus selectivity, a robust information processing operation in which cortical neurons respond more to some stimuli than to others. Previous models assume that these neurons receive the highest weighted input from an ensemble encoding the preferred stimulus, but dendrites enable other possibilities. Nonlinear dendritic processing can produce stimulus selectivity based on the spatial distribution of synapses, even if the total preferred stimulus weight does not exceed that of nonpreferred stimuli. Using a multi-subunit nonlinear model, we demonstrate that stimulus selectivity can arise from the spatial distribution of synapses. We propose this as a general mechanism for information processing by neurons possessing dendritic trees. Moreover, we show that this implementation of stimulus selectivity increases the neuron's robustness to synaptic and dendritic failure. Importantly, our model can maintain stimulus selectivity for a larger range of loss of synapses or dendrites than an equivalent linear model. We then use a layer 2/3 biophysical neuron model to show that our implementation is consistent with two recent experimental observations: (1) one can observe a mixture of selectivities in dendrites that can differ from the somatic selectivity, and (2) hyperpolarization can broaden somatic tuning without affecting

dendritic tuning. Our model predicts that an initially nonselective neuron can become selective when depolarized. In addition to motivating new experiments, the model's increased robustness to synapses and dendrites loss provides a starting point for fault-resistant neuromorphic chip development.

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86. Recurrent Information Optimization with Local, Metaplastic Synaptic Dynamics

Sensen Liu, ShiNung Ching

https://doi.org/10.1162/neco_a_00993 | pp. 2528-2552

We consider the problem of optimizing information-theoretic quantities in recurrent networks via synaptic learning. In contrast to feedforward networks, the recurrence presents a key challenge insofar as an optimal learning rule must aggregate the joint distribution of the whole network. This challenge, in particular, makes a local policy (i.e., one that depends on only pairwise interactions) difficult. Here, we report a local metaplastic learning rule that performs approximate optimization by estimating whole-network statistics through the use of several slow, nested dynamical variables. These dynamics provide the rule with both anti-Hebbian and Hebbian components, thus allowing for decorrelating and correlating learning regimes that can occur when either is favorable for optimality. We demonstrate the performance of the synthesized rule in comparison to classical BCM dynamics and use the networks to conduct history-dependent tasks that highlight the advantages of recurrence. Finally, we show the consistency of the resultant learned networks with notions of criticality, including balanced ratios of excitation and inhibition.

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87. Nonnegative Matrix Factorization with Rank Regularization and Hard Constraint

Ronghua Shang, Chiyang Liu, Meng Yang, Licheng Jiao, Rustam Stolkin

https://doi.org/10.1162/neco_a_00995 | pp. 2553-2579

Nonnegative matrix factorization (NMF) is well known to be an effective tool for dimensionality reduction in problems involving big data. For this reason, it frequently appears in many areas of scientific and engineering literature. This letter proposes a novel semisupervised NMF algorithm for overcoming a variety of problems associated with NMF algorithms, including poor use of prior information, negative impact on manifold structure of the sparse constraint, and inaccurate graph construction. Our proposed algorithm, nonnegative matrix factorization with rank regularization and hard constraint (NMFRC), incorporates label information into data representation as a hard constraint, which makes full use of prior information. NMFRC also measures pairwise similarity according to geodesic distance rather than Euclidean distance. This results in more accurate measurement of pairwise relationships, resulting in more effective manifold information. Furthermore, NMFRC adopts rank constraint instead of norm constraints for regularization to balance the sparseness and smoothness of data. In this way, the new data representation is more representative and has better interpretability. Experiments on real data sets suggest that NMFRC outperforms four other state-of-the-art algorithms in terms of clustering accuracy.

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88. Differential Covariance: A New Class of Methods to Estimate Sparse Connectivity from Neural Recordings

Tiger W. Lin, Anup Das, Giri P. Krishnan, Maxim Bazhenov, Terrence J. Sejnowski

https://doi.org/10.1162/neco_a_01008 | pp. 2581-2632

With our ability to record more neurons simultaneously, making sense of these data is a challenge. Functional connectivity is one popular way to study the relationship of multiple neural signals. Correlation-based methods are a set of currently well-used techniques for functional connectivity estimation. However, due to explaining away and unobserved common inputs (Stevenson, Rebesco, Miller, & Körding, 2008), they produce spurious connections. The general linear model (GLM), which models spike trains as Poisson processes (Okatan, Wilson, & Brown, 2005 ; Truccolo, Eden, Fellows, Donoghue, & Brown, 2005 ; Pillow et al., 2008), avoids these confounds. We develop here a new class of methods by using differential signals based on simulated intracellular voltage recordings. It is equivalent to a regularized AR(2) model. We also expand the method to simulated local field potential recordings and calcium imaging. In all of our simulated data, the differential covariance-based methods achieved performance better than or similar to the GLM method and required fewer data samples. This new class of methods provides alternative ways to analyze neural signals.

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89. Active Inference, Curiosity and Insight

Karl Friston, Marco Lin, Chris Frith, Giovanni Pezzulo, J. Allan Hobson, Sasha Ondobaka

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This article offers a formal account of curiosity and insight in terms of active (Bayesian) inference. It deals with the dual problem of inferring states of the world and learning its statistical structure. In contrast to current trends in machine learning (e.g., deep learning), we focus on how people attain insight and understanding using just a handful of observations, which are solicited through curious behavior. We use simulations of abstract rule learning and approximate Bayesian inference to show that minimizing (expected) variational free energy leads to active sampling of novel contingencies. This epistemic behavior closes explanatory gaps in generative models of the world, thereby reducing uncertainty and satisfying curiosity. We then move from epistemic learning to model selection or structure learning to show how abductive processes emerge when agents test plausible hypotheses about symmetries (i.e., invariances or rules) in their generative models. The ensuing Bayesian model reduction evinces mechanisms associated with sleep and has all the hallmarks of "aha" moments. This formulation moves toward a computational account of consciousness in the pre-Cartesian sense of sharable knowledge (i.e., con: "together"; scire: "to know").

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90. Memory States and Transitions between Them in Attractor Neural Networks

Stefano Recanatesi, Mikhail Katkov, Misha Tsodyks

https://doi.org/10.1162/neco_a_00998 | pp. 2684-2711

Human memory is capable of retrieving similar memories to a just retrieved one. This associative ability is at the base of our everyday processing of information. Current models of memory have not been able to underpin the mechanism that the brain could use in order to actively exploit similarities between memories.

The current idea is that to induce transitions in attractor neural networks, it is necessary to extinguish the current memory. We introduce a novel mechanism capable of inducing transitions between memories where similarities between memories are actively exploited by the neural dynamics to retrieve a new memory. Populations of neurons that are selective for multiple memories play a crucial role in this mechanism by becoming attractors on their own. The mechanism is based on the ability of the neural network to control the excitation-inhibition balance.

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91. Infinite von Mises–Fisher Mixture Modeling of Whole Brain fMRI Data

Rasmus Røge, Kristoffer H. Madsen, Mikkel N. Schmidt, Morten Mørup

https://doi.org/10.1162/neco_a_01000 | pp. 2712-2741

Cluster analysis of functional magnetic resonance imaging (fMRI) data is often performed using gaussian mixture models, but when the time series are standardized such that the data reside on a hypersphere, this modeling assumption is questionable. The consequences of ignoring the underlying spherical manifold are rarely analyzed, in part due to the computational challenges imposed by directional statistics. In this letter, we discuss a Bayesian von Mises-Fisher (vMF) mixture model for data on the unit hypersphere and present an efficient inference procedure based on collapsed Markov chain Monte Carlo sampling. Comparing the vMF and gaussian mixture models on synthetic data, we demonstrate that the vMF model has a slight advantage inferring the true underlying clustering when compared to gaussian-based models on data generated from both a mixture of vMFs and a mixture of gaussians subsequently normalized. Thus, when performing model selection, the two models are not in agreement. Analyzing multisubject whole brain resting-state fMRI data from healthy adult subjects, we find that the vMF mixture model is considerably more reliable than the gaussian mixture model when comparing solutions across models trained on different groups of subjects, and again we find that the two models disagree on the optimal number of components. The analysis indicates that the fMRI data support more than a thousand clusters, and we confirm this is not a result of overfitting by demonstrating better prediction on data from held-out subjects. Our results highlight the utility of using directional statistics to model standardized fMRI data and demonstrate that whole brain segmentation of fMRI data requires a very large number of functional units in order to adequately account for the discernible statistical patterns in the data.

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92. Toward an Open-Ended BCI: A User-Centered Coadaptive Design

Kiret Dhindsa, Dean Carcone, Suzanna Becker

https://doi.org/10.1162/neco_a_01001 | pp. 2742-2768

Brain-computer interfaces (BCIs) allow users to control a device by interpreting their brain activity. For simplicity, these devices are designed to be operated by purposefully modulating specific predetermined neurophysiological signals, such as the sensorimotor rhythm. However, the ability to modulate a given neurophysiological signal is highly variable across individuals, contributing to the inconsistent performance of BCIs for different users. These differences suggest that individuals who experience poor BCI performance with one class of brain signals might have good results with another. In order to take advantage of individual abilities as they relate to BCI control, we need to move beyond the current approaches. In this letter, we explore a new BCI design aimed at a more individualized and user-focused experience, which we call open-ended BCI. Individual users were given the freedom to discover their own mental strategies as

opposed to being trained to modulate a given brain signal. They then underwent multiple coadaptive training sessions with the BCI. Our first open-ended BCI performed similarly to comparable BCIs while accommodating a wider variety of mental strategies without a priori knowledge of the specific brain signals any individual might use. Post hoc analysis revealed individual differences in terms of which sensory modality yielded optimal performance. We found a large and significant effect of individual differences in background training and expertise, such as in musical training, on BCI performance. Future research should be focused on finding more generalized solutions to user training and brain state decoding methods to fully utilize the abilities of different individuals in an open-ended BCI. Accounting for each individual's areas of expertise could have important implications on BCI training and BCI application design.

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93. The Two-Dimensional Gabor Function Adapted to Natural Image Statistics: A Model of Simple-Cell Receptive Fields and Sparse Structure in Images

P. N. Loxley

https://doi.org/10.1162/neco_a_00997 | pp. 2769-2799

The two-dimensional Gabor function is adapted to natural image statistics, leading to a tractable probabilistic generative model that can be used to model simple cell receptive field profiles, or generate basis functions for sparse coding applications. Learning is found to be most pronounced in three Gabor function parameters representing the size and spatial frequency of the two-dimensional Gabor function and characterized by a nonuniform probability distribution with heavy tails. All three parameters are found to be strongly correlated, resulting in a basis of multiscale Gabor functions with similar aspect ratios and size-dependent spatial frequencies. A key finding is that the distribution of receptive-field sizes is scale invariant over a wide range of values, so there is no characteristic receptive field size selected by natural image statistics. The Gabor function aspect ratio is found to be approximately conserved by the learning rules and is therefore not well determined by natural image statistics. This allows for three distinct solutions: a basis of Gabor functions with sharp orientation resolution at the expense of spatial-frequency resolution, a basis of Gabor functions with sharp spatial-frequency resolution at the expense of orientation resolution, or a basis with unit aspect ratio. Arbitrary mixtures of all three cases are also possible. Two parameters controlling the shape of the marginal distributions in a probabilistic generative model fully account for all three solutions. The best-performing probabilistic generative model for sparse coding applications is found to be a gaussian copula with Pareto marginal probability density functions.

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94. Nonconvex Policy Search Using Variational Inequalities

Yusen Zhan, Haitham Bou Ammar, Matthew E. Taylor

https://doi.org/10.1162/neco_a_01004 | pp. 2800-2824

Policy search is a class of reinforcement learning algorithms for finding optimal policies in control problems with limited feedback. These methods have been shown to be successful in high-dimensional problems such as robotics control. Though successful, current methods can lead to unsafe policy parameters that potentially could damage hardware units. Motivated by such constraints, we propose projection-based methods for safe policies. These methods, however, can handle only convex policy constraints. In this letter, we propose the first safe policy search reinforcement learner capable of operating under nonconvex policy

constraints. This is achieved by observing, for the first time, a connection between nonconvex variational inequalities and policy search problems. We provide two algorithms, Mann and two-step iteration, to solve the above problems and prove convergence in the nonconvex stochastic setting. Finally, we demonstrate the performance of the algorithms on six benchmark dynamical systems and show that our new method is capable of outperforming previous methods under a variety of settings.

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95. Constrained ERM Learning of Canonical Correlation Analysis: A Least Squares Perspective

Jia Cai, Hongwei Sun

https://doi.org/10.1162/neco_a_00996 | pp. 2825-2859

Canonical correlation analysis (CCA) is a useful tool in detecting the latent relationship between two sets of multivariate variables. In theoretical analysis of CCA, a regularization technique is utilized to investigate the consistency of its analysis. This letter addresses the consistency property of CCA from a least squares view. We construct a constrained empirical risk minimization framework of CCA and apply a two-stage randomized Kaczmarz method to solve it. In the first stage, we remove the noise, and in the second stage, we compute the canonical weight vectors. Rigorous theoretical consistency is addressed. The statistical consistency of this novel scenario is extended to the kernel version of it. Moreover, experiments on both synthetic and real-world data sets demonstrate the effectiveness and efficiency of the proposed algorithms.

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96. Rat Prefrontal Cortex Inactivations during Decision Making Are Explained by Bistable Attractor Dynamics

Alex T. Piet, Jeffrey C. Erlich, Charles D. Kopec, Carlos D. Brody

https://doi.org/10.1162/neco_a_01005 | pp. 2861-2886

Two-node attractor networks are flexible models for neural activity during decision making. Depending on the network configuration, these networks can model distinct aspects of decisions including evidence integration, evidence categorization, and decision memory. Here, we use attractor networks to model recent causal perturbations of the frontal orienting fields (FOF) in rat cortex during a perceptual decision-making task (Erlich, Brunton, Duan, Hanks, & Brody, 2015). We focus on a striking feature of the perturbation results. Pharmacological silencing of the FOF resulted in a stimulus-independent bias. We fit several models to test whether integration, categorization, or decision memory could account for this bias and found that only the memory configuration successfully accounts for it. This memory model naturally accounts for optogenetic perturbations of FOF in the same task and correctly predicts a memory-duration-dependent deficit caused by silencing FOF in a different task. Our results provide mechanistic support for a "postcategorization" memory role of the FOF in upcoming choices.

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97. Simultaneous Estimation of Nongaussian Components and Their Correlation Structure

Hiroaki Sasaki, Michael U. Gutmann, Hayaru Shouno, Aapo Hyvärinen

https://doi.org/10.1162/neco_a_01006 | pp. 2887-2924

The statistical dependencies that independent component analysis (ICA) cannot remove often provide rich information beyond the linear independent components. It would thus be very useful to estimate the dependency structure from data. While such models have been proposed, they have usually concentrated on higher-order correlations such as energy (square) correlations. Yet linear correlations are a fundamental and informative form of dependency in many real data sets. Linear correlations are usually completely removed by ICA and related methods so they can only be analyzed by developing new methods that explicitly allow for linearly correlated components. In this article, we propose a probabilistic model of linear nongaussian components that are allowed to have both linear and energy correlations. The precision matrix of the linear components is assumed to be randomly generated by a higher-order process and explicitly parameterized by a parameter matrix. The estimation of the parameter matrix is shown to be particularly simple because using score-matching (Hyvärinen, 2005), the objective function is a quadratic form. Using simulations with artificial data, we demonstrate that the proposed method improves the identifiability of nongaussian components by simultaneously learning their correlation structure. Applications on simulated complex cells with natural image input, as well as spectrograms of natural audio data, show that the method finds new kinds of dependencies between the components.

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98. Blind Nonnegative Source Separation Using Biological Neural Networks

Cengiz Pehlevan, Sreyas Mohan, Dmitri B. Chklovskii

https://doi.org/10.1162/neco_a_01007 | pp. 2925-2954

Blind source separation-the extraction of independent sources from a mixture-is an important problem for both artificial and natural signal processing. Here, we address a special case of this problem when sources (but not the mixing matrix) are known to be nonnegative-for example, due to the physical nature of the sources. We search for the solution to this problem that can be implemented using biologically plausible neural networks. Specifically, we consider the online setting where the data set is streamed to a neural network. The novelty of our approach is that we formulate blind nonnegative source separation as a similarity matching problem and derive neural networks from the similarity matching objective. Importantly, synaptic weights in our networks are updated according to biologically plausible local learning rules.

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99. Effect of Morphologic Features of Neurons on the Extracellular Electric Potential: A Simulation Study Using Cable Theory and Electro-Quasi-Static Equations

Robert Bestel, Revathi Appali, Ursula van Rienen, Christiane Thielemann

https://doi.org/10.1162/neco_a_01019 | pp. 2955-2978

Microelectrode arrays serve as an indispensable tool in electro-physiological research to study the electrical activity of neural cells, enabling measurements of single cell as well as network communication analysis. Recent experimental studies have reported that the neuronal geometry has an influence on electrical signaling and extracellular recordings. However, the corresponding mechanisms are not yet fully understood and require further investigation. Allowing systematic parameter studies, computational modeling provides the opportunity to examine the underlying effects that influence extracellular potentials. In this letter, we present an in silico single cell model to analyze the effect of geometrical variability on the extracellular electric potentials. We describe finite element models of a single neuron with varying geometric complexity

in three-dimensional space. The electric potential generation of the neuron is modeled using Hodgkin-Huxley equations. The signal propagation is described with electro-quasi-static equations, and results are compared with corresponding cable equation descriptions. Our results show that both the geometric dimensions and the distribution of ion channels of a neuron are critical factors that significantly influence both the amplitude and shape of extracellular potentials.

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100. Discrete Sparse Coding

Georgios Exarchakis, Jörg Lücke

https://doi.org/10.1162/neco_a_01015 | pp. 2979-3013

Sparse coding algorithms with continuous latent variables have been the subject of a large number of studies. However, discrete latent spaces for sparse coding have been largely ignored. In this work, we study sparse coding with latents described by discrete instead of continuous prior distributions. We consider the general case in which the latents (while being sparse) can take on any value of a finite set of possible values and in which we learn the prior probability of any value from data. This approach can be applied to any data generated by discrete causes, and it can be applied as an approximation of continuous causes. As the prior probabilities are learned, the approach then allows for estimating the prior shape without assuming specific functional forms. To efficiently train the parameters of our probabilistic generative model, we apply a truncated expectation-maximization approach (expectation truncation) that we modify to work with a general discrete prior. We evaluate the performance of the algorithm by applying it to a variety of tasks: (1) we use artificial data to verify that the algorithm can recover the generating parameters from a random initialization, (2) use image patches of natural images and discuss the role of the prior for the extraction of image components, (3) use extracellular recordings of neurons to present a novel method of analysis for spiking neurons that includes an intuitive discretization strategy, and (4) apply the algorithm on the task of encoding audio waveforms of human speech. The diverse set of numerical experiments presented in this letter suggests that discrete sparse coding algorithms can scale efficiently to work with realistic data sets and provide novel statistical quantities to describe the structure of the data.

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101. A Robust Regression Framework with Laplace Kernel-Induced Loss

Liming Yang, Zhuo Ren, Yidan Wang, Hongwei Dong

https://doi.org/10.1162/neco_a_01002 | pp. 3014-3039

This work proposes a robust regression framework with nonconvex loss function. Two regression formulations are presented based on the Laplace kernel-induced loss (LK-loss). Moreover, we illustrate that the LK-loss function is a nice approximation for the zero-norm. However, nonconvexity of the LK-loss makes it difficult to optimize. A continuous optimization method is developed to solve the proposed framework. The problems are formulated as DC (difference of convex functions) programming. The corresponding DC algorithms (DCAs) converge linearly. Furthermore, the proposed algorithms are applied directly to determine the hardness of licorice seeds using near-infrared spectral data with noisy input. Experiments in eight spectral regions show that the proposed methods improve generalization compared with the traditional support vector regressions (SVR), especially in high-frequency regions. Experiments on several benchmark data sets demonstrate that the proposed methods achieve better results than the traditional regression methods in most of data sets we have considered.

102. Sparse Covariance Matrix Estimation by DCA-Based Algorithms

Duy Nhat Phan, Hoai An Le Thi, Tao Pham Dinh

https://doi.org/10.1162/neco_a_01012 | pp. 3040-3077

This letter proposes a novel approach using the [Formula: see text]-norm regularization for the sparse covariance matrix estimation (SCME) problem. The objective function of SCME problem is composed of a nonconvex part and the [Formula: see text] term, which is discontinuous and difficult to tackle. Appropriate DC (difference of convex functions) approximations of [Formula: see text]-norm are used that result in approximation SCME problems that are still nonconvex. DC programming and DCA (DC algorithm), powerful tools in nonconvex programming framework, are investigated. Two DC formulations are proposed and corresponding DCA schemes developed. Two applications of the SCME problem that are considered are classification via sparse quadratic discriminant analysis and portfolio optimization. A careful empirical experiment is performed through simulated and real data sets to study the performance of the proposed algorithms. Numerical results showed their efficiency and their superiority compared with seven state-of-the-art methods.

103. Margin Error Bounds for Support Vector Machines on Reproducing Kernel Banach Spaces

Liangzhi Chen, Haizhang Zhang

https://doi.org/10.1162/neco_a_01013 | pp. 3078-3093

Support vector machines, which maximize the margin from patterns to the separation hyperplane subject to correct classification, have received remarkable success in machine learning. Margin error bounds based on Hilbert spaces have been introduced in the literature to justify the strategy of maximizing the margin in SVM. Recently, there has been much interest in developing Banach space methods for machine learning. Large margin classification in Banach spaces is a focus of such attempts. In this letter we establish a margin error bound for the SVM on reproducing kernel Banach spaces, thus supplying statistical justification for large-margin classification in Banach spaces.

104. An Initialization Method Based on Hybrid Distance for k-Means Algorithm

Jie Yang, Yan Ma, Xiangfen Zhang, Shunbao Li, Yuping Zhang

https://doi.org/10.1162/neco_a_01014 | pp. 3094-3117

The traditional [Formula: see text]-means algorithm has been widely used as a simple and efficient clustering method. However, the performance of this algorithm is highly dependent on the selection of initial cluster centers. Therefore, the method adopted for choosing initial cluster centers is extremely important. In this letter, we redefine the density of points according to the number of its neighbors, as well as the distance between points and their neighbors. In addition, we define a new distance measure that considers both Euclidean distance and density. Based on that, we propose an algorithm for selecting initial cluster centers that can dynamically adjust the weighting parameter. Furthermore, we propose a new internal clustering validation measure, the clustering validation index based on the neighbors (CVN), which can be exploited to

select the optimal result among multiple clustering results. Experimental results show that the proposed algorithm outperforms existing initialization methods on real-world data sets and demonstrates the adaptability of the proposed algorithm to data sets with various characteristics.

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105. Noise-Robust Modes of the Retinal Population Code Have the Geometry of “Ridges” and Correspond to Neuronal Communities

Adrianna Loback, Jason Prentice, Mark L. Ioffe, Michael J. Berry

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An appealing new principle for neural population codes is that correlations among neurons organize neural activity patterns into a discrete set of clusters, which can each be viewed as a noise-robust population codeword. Previous studies assumed that these codewords corresponded geometrically with local peaks in the probability landscape of neural population responses. Here, we analyze multiple data sets of the responses of approximately 150 retinal ganglion cells and show that local probability peaks are absent under broad, nonrepeated stimulus ensembles, which are characteristic of natural behavior. However, we find that neural activity still forms noise-robust clusters in this regime, albeit clusters with a different geometry. We start by defining a soft local maximum, which is a local probability maximum when constrained to a fixed spike count. Next, we show that soft local maxima are robustly present and can, moreover, be linked across different spike count levels in the probability landscape to form a ridge. We found that these ridges comprise combinations of spiking and silence in the neural population such that all of the spiking neurons are members of the same neuronal community, a notion from network theory. We argue that a neuronal community shares many of the properties of Donald Hebb's classic cell assembly and show that a simple, biologically plausible decoding algorithm can recognize the presence of a specific neuronal community.

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106. Delay Differential Analysis of Seizures in Multichannel Electrocorticography Data

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High-density electrocorticogram (ECoG) electrodes are capable of recording neurophysiological data with high temporal resolution with wide spatial coverage. These recordings are a window to understanding how the human brain processes information and subsequently behaves in healthy and pathologic states. Here, we describe and implement delay differential analysis (DDA) for the characterization of ECoG data obtained from human patients with intractable epilepsy. DDA is a time-domain analysis framework based on embedding theory in nonlinear dynamics that reveals the nonlinear invariant properties of an unknown dynamical system. The DDA embedding serves as a low-dimensional nonlinear dynamical basis onto which the data are mapped. This greatly reduces the risk of overfitting and improves the method's ability to fit classes of data. Since the basis is built on the dynamical structure of the data, preprocessing of the data (e.g., filtering) is not necessary. We performed a large-scale search for a DDA model that best fit ECoG recordings using a genetic algorithm to qualitatively discriminate between different cortical states and epileptic events for a set of 13 patients. A single DDA model with only three polynomial terms was identified. Singular value decomposition across the feature space of the model revealed both global and local dynamics that could differentiate electrographic and electroclinical seizures and provided insights into highly

localized seizure onsets and diffuse seizure terminations. Other common ECoG features such as interictal periods, artifacts, and exogenous stimuli were also analyzed with DDA. This novel framework for signal processing of seizure information demonstrates an ability to reveal unique characteristics of the underlying dynamics of the seizure and may be useful in better understanding, detecting, and maybe even predicting seizures.

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107. First Passage Time Memory Lifetimes for Simple, Multistate Synapses

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Memory models based on synapses with discrete and bounded strengths store new memories by forgetting old ones. Memory lifetimes in such memory systems may be defined in a variety of ways. A mean first passage time (MFPT) definition overcomes much of the arbitrariness and many of the problems associated with the more usual signal-to-noise ratio (SNR) definition. We have previously computed MFPT lifetimes for simple, binary-strength synapses that lack internal, plasticity-related states. In simulation we have also seen that for multistate synapses, optimality conditions based on SNR lifetimes are absent with MFPT lifetimes, suggesting that such conditions may be artifactual. Here we extend our earlier work by computing the entire first passage time (FPT) distribution for simple, multistate synapses, from which all statistics, including the MFPT lifetime, may be extracted. For this, we develop a Fokker-Planck equation using the jump moments for perceptron activation. Two models are considered that satisfy a particular eigenvector condition that this approach requires. In these models, MFPT lifetimes do not exhibit optimality conditions, while in one but not the other, SNR lifetimes do exhibit optimality. Thus, not only are such optimality conditions artifacts of the SNR approach, but they are also strongly model dependent. By examining the variance in the FPT distribution, we may identify regions in which memory storage is subject to high variability, although MFPT lifetimes are nevertheless robustly positive. In such regions, SNR lifetimes are typically (defined to be) zero. FPT-defined memory lifetimes therefore provide an analytically superior approach and also have the virtue of being directly related to a neuron's firing properties.

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108. Capturing the Dynamical Repertoire of Single Neurons with Generalized Linear Models

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A key problem in computational neuroscience is to find simple, tractable models that are nevertheless flexible enough to capture the response properties of real neurons. Here we examine the capabilities of recurrent point process models known as Poisson generalized linear models (GLMs). These models are defined by a set of linear filters and a point nonlinearity and are conditionally Poisson spiking. They have desirable statistical properties for fitting and have been widely used to analyze spike trains from electrophysiological recordings. However, the dynamical repertoire of GLMs has not been systematically compared to that of real neurons. Here we show that GLMs can reproduce a comprehensive suite of canonical neural response behaviors, including tonic and phasic spiking, bursting, spike rate adaptation, type I and type II excitation, and two forms of bistability. GLMs can also capture stimulus-dependent changes in spike timing precision and reliability that mimic those observed in real neurons, and can exhibit

varying degrees of stochasticity, from virtually deterministic responses to greater-than-Poisson variability. These results show that Poisson GLMs can exhibit a wide range of dynamic spiking behaviors found in real neurons, making them well suited for qualitative dynamical as well as quantitative statistical studies of single-neuron and population response properties.

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109. Neural Decoding: A Predictive Viewpoint

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Decoding in the context of brain-machine interface is a prediction problem, with the aim of retrieving the most accurate kinematic predictions attainable from the available neural signals. While selecting models that reduce the prediction error is done to various degrees, decoding has not received the attention that the fields of statistics and machine learning have lavished on the prediction problem in the past two decades. Here, we take a more systematic approach to the decoding prediction problem and search for risk-optimized reverse regression, optimal linear estimation (OLE), and Kalman filter models within a large model space composed of several nonlinear transformations of neural spike counts at multiple temporal lags. The reverse regression decoding framework is a standard prediction problem, where penalized methods such as ridge regression or Lasso are routinely used to find minimum risk models. We argue that minimum risk reverse regression is always more efficient than OLE and also happens to be 44% more efficient than a standard Kalman filter in a particular application of offline reconstruction of arm reaches of a rhesus macaque monkey. Yet model selection for tuning curves-based decoding models such as OLE and Kalman filtering is not a standard statistical prediction problem, and no efficient method exists to identify minimum risk models. We apply several methods to build low-risk models and show that in our application, a Kalman filter that includes multiple carefully chosen observation equations per neural unit is 67% more efficient than a standard Kalman filter, but with the drawback that finding such a model is computationally very costly.

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110. Dopamine, Inference, and Uncertainty

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The hypothesis that the phasic dopamine response reports a reward prediction error has become deeply entrenched. However, dopamine neurons exhibit several notable deviations from this hypothesis. A coherent explanation for these deviations can be obtained by analyzing the dopamine response in terms of Bayesian reinforcement learning. The key idea is that prediction errors are modulated by probabilistic beliefs about the relationship between cues and outcomes, updated through Bayesian inference. This account can explain dopamine responses to inferred value in sensory preconditioning, the effects of cue preexposure (latent inhibition), and adaptive coding of prediction errors when rewards vary across orders of magnitude. We further postulate that orbitofrontal cortex transforms the stimulus representation through recurrent dynamics, such that a simple error-driven learning rule operating on the transformed representation can implement the Bayesian reinforcement learning update.

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111. Learning Simpler Language Models with the Differential State Framework

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Learning useful information across long time lags is a critical and difficult problem for temporal neural models in tasks such as language modeling. Existing architectures that address the issue are often complex and costly to train. The differential state framework (DSF) is a simple and high-performing design that unifies previously introduced gated neural models. DSF models maintain longer-term memory by learning to interpolate between a fast-changing data-driven representation and a slowly changing, implicitly stable state. Within the DSF framework, a new architecture is presented, the delta-RNN. This model requires hardly any more parameters than a classical, simple recurrent network. In language modeling at the word and character levels, the delta-RNN outperforms popular complex architectures, such as the long short-term memory (LSTM) and the gated recurrent unit (GRU), and, when regularized, performs comparably to several state-of-the-art baselines. At the subword level, the delta-RNN's performance is comparable to that of complex gated architectures.

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112. Learning Rates for Classification with Gaussian Kernels

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This letter aims at refined error analysis for binary classification using support vector machine (SVM) with gaussian kernel and convex loss. Our first result shows that for some loss functions, such as the truncated quadratic loss and quadratic loss, SVM with gaussian kernel can reach the almost optimal learning rate provided the regression function is smooth. Our second result shows that for a large number of loss functions, under some Tsybakov noise assumption, if the regression function is infinitely smooth, then SVM with gaussian kernel can achieve the learning rate of order [Formula: see text], where [Formula: see text] is the number of samples.

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113. Refined Spectral Clustering via Embedded Label Propagation

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Spectral clustering is a key research topic in the field of machine learning and data mining. Most of the existing spectral clustering algorithms are built on gaussian Laplacian matrices, which is sensitive to parameters. We propose a novel parameter-free distance-consistent locally linear embedding. The proposed distance-consistent LLE can promise that edges between closer data points are heavier. We also propose a novel improved spectral clustering via embedded label propagation. Our algorithm is built on two advancements of the state of the art. First is label propagation, which propagates a node's labels to neighboring nodes according to their proximity. We perform standard spectral clustering on original data and assign each cluster with [Formula: see text]-nearest data points and then we propagate labels through dense unlabeled data regions. Second is manifold learning, which has been widely used for its capacity to leverage the manifold structure of data points. Extensive experiments on various data sets validate the

superiority of the proposed algorithm compared to state-of-the-art spectral algorithms.